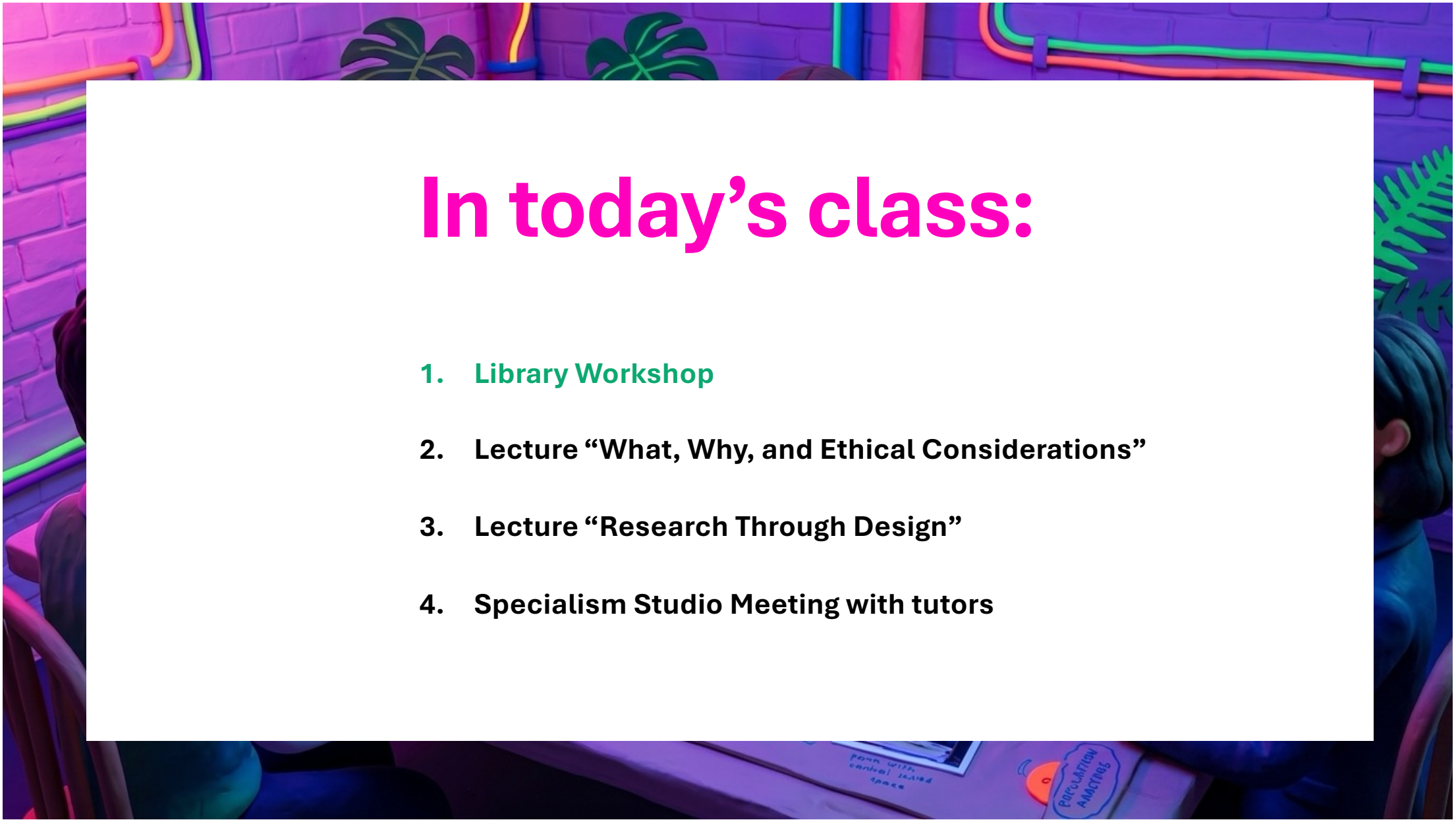


SD5008
Research and Analysis for Design

WEEK 02

Jen Yoohyun LEE / Henry Rui HE / Mohana DAS / Bharati DAS



In today’s class:

1. Library Workshop
2. Lecture “What, Why, and Ethical Considerations”
3. Lecture “Research Through Design”
4. Specialism Studio Meeting with tutors

Schedule

Week	Date & Place	Topic	Milestones
1	03/09/2026 TU201	**[Lecture]** Introduction to Design Research **[Special Lecture]** Prof. Jaden Park **[Tutorial]**	- Group arrangement after lecture
2	10/09/2026 V322	**[Library Workshop]** **[Lecture]** What & Why: Identifying and defining research dimensions; Research Through Design **[Tutorial]**	- Initial project idea / topic - Research purpose - Begin background & precedent review
3	17/09/2026 TU201	**[Lecture]** How #1: Approaches of Qualitative Research & Quantitative Research **[Tutorial]**	- Continue background review - Draft research question / creative proposition
4	24/09/2026 TU201	**[Lecture]** How #2: Data gathering mechanisms + AI **[Tutorial]**	-Submit **Research study project presentation 1** + Research topic + Literature mapping + Inquisitive direction + Research modalities in consideration
-	01/10/2026	*No Class*	- Initiate data gathering process



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What: Research Topic

- A research topic is the broad subject or field of interest to investigate
- It answers what general area to explore or what problem needs attention
- It can be very broad and comprehensive, spanning multiple sub-themes and disciplines
- For an example, “AI in education”



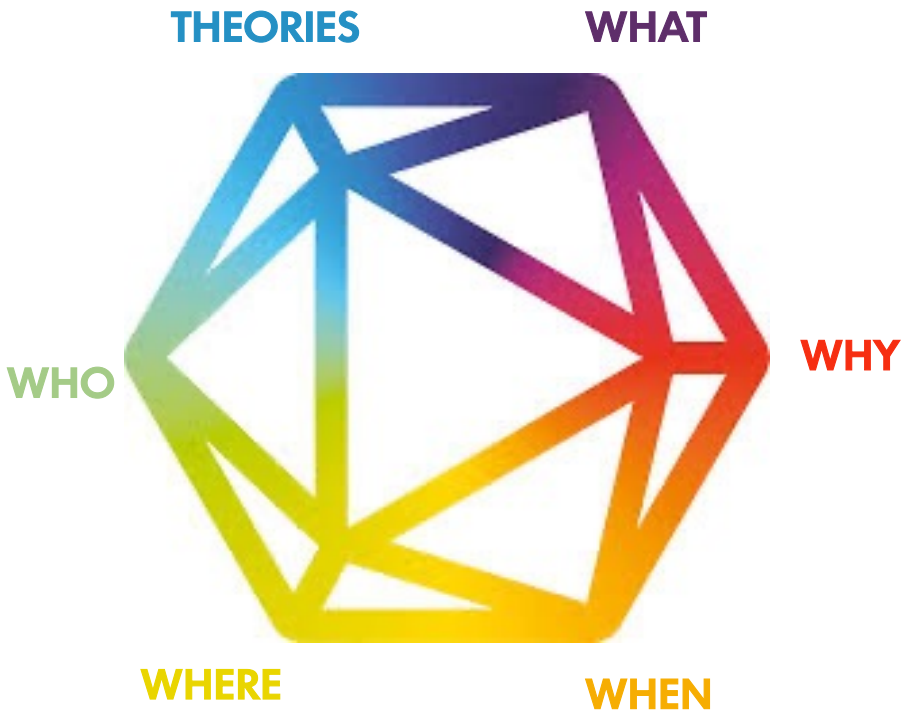
What: Research Scope

- Research scope is the chosen focus area
- It defines the operational boundaries and extent of a study, narrowing down to specific objectives, target population, variables, methods, and limitations
- It helps us to focus and provides a clear understanding of what will be studied
    - It clarifies why specific data will be collected
    - It defines an end point for a research
- For an example, "A survey of AI chatbot usage among 200 undergraduate computer science students in Hong Kong during the Fall 2026 semester"



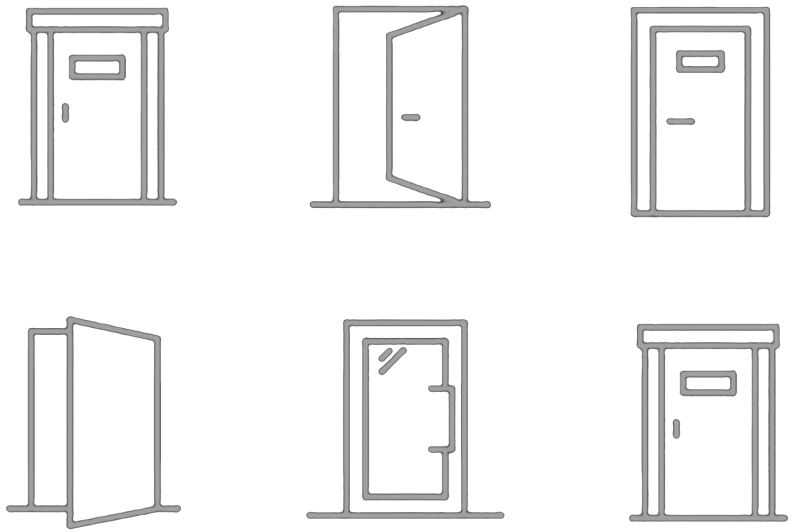
What: Research Scope

- Identifying components for conducting research
  - ✓ WHAT – what is the research topic?
  - ✓ WHY – what is the overall goal of the research?
  - ✓ WHEN – how much time is available? when is the research taking place?
  - ✓ WHERE – where is the research located?
  - ✓ WHO – who are the target population?
  - ✓ THEORIES – what are the key concepts that determine the perspective?



What: Research Question

- A research question is a question that a study or research project aims to answer
- For design research, having an open-ended research question would allow you to explore different possibilities
  - ❑ Open-ended questions are not black or white
  - ❑ Closed-ended questions are used to test hypotheses, whether they are right or wrong
- A research question serves as a guiding framework for research



What: Research Question

- A research question addresses an issue or a problem
- A research question should define exactly what you are trying to find out
  - ❑ What …? : to identify …
  - ❑ Why …? : to explain …
  - ❑ How …? : to explore …
- Identifying a research question has a domino effect on the rest of the study
- The research question influences research methodology, studied demographic, data collection, and data analysis



What & Why: Research Background

Background Review – Administrative Approach

- The purpose is to map out the broad overview context, real-world setting, and practical relevance of the research problem
- It covers definitions, historical context, policy importance, and basic descriptive overviews
- It is generally descriptive, general-to-specific, and straightforward, without deep academic debate
- It answers why the topic matters and justifies the practical need for the study.

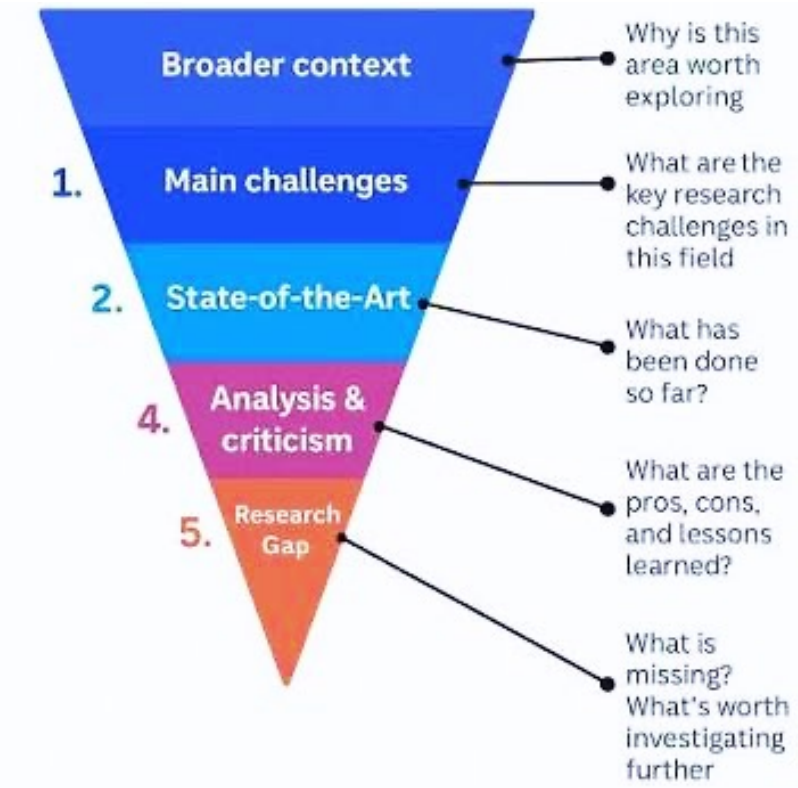


https://www.magnific.com/free-photos-vectors/critical-review

What & Why: Research Background

Literature Review – Academic Approach

- It *critically* summarizes *prior* academic research and theoretical frameworks
- It is analytical, general-to-specific, scholarly, heavily cited, and organized by themes or variables
- It shares the results of other studies that are closely related to the one being undertaken
- It relates a study to the larger, ongoing dialogue in the literature, filling in gaps and extending prior studies
- It identifies **gaps** in current knowledge.



https://www.facebook.com/groups/researchscholarsalerts/posts/2475589809500626/

Literature Search

- A literature search is a planned and organized way to find published books, articles, and reports on a specific topic.
- Keywords should be selected from the research scope and research question(s) –synonyms or different names for the main ideas should be sought.
- Rather than simply searching on generic search engines, use key academic databases such as Google Scholar, JSTOR, PubMed, Web of Science, Scopus, etc.
- Use keywords and Boolean operators (AND, OR, NOT) to refine searches.



https://iziel.com/literature-search-study/

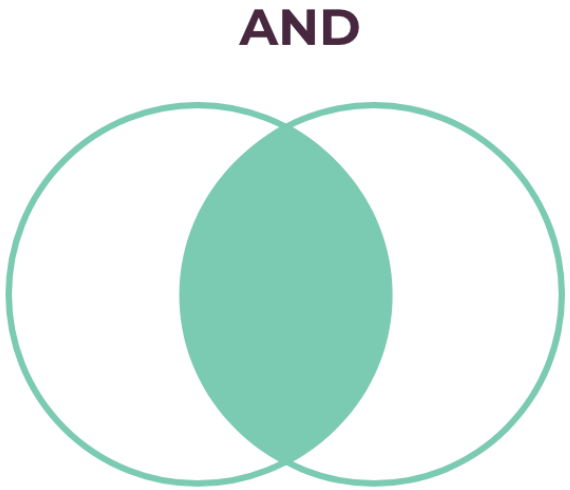
Literature Search

How to use Boolean Operators

1. Using “AND”

- Purpose: To *narrow* your search by including only sources that contain both terms.
- Search Query: “elderly care” AND “aging in place”
- Explanation: This search will return results that include both “elderly care” and “aging in place.”

→ Use this when investigating how aging in place practices specifically drive elderly care, filtering out general papers about elderly care that do not address aging in place.



https://recruitinginnovation.com/understanding-boolean-search-and-boolean-operators/

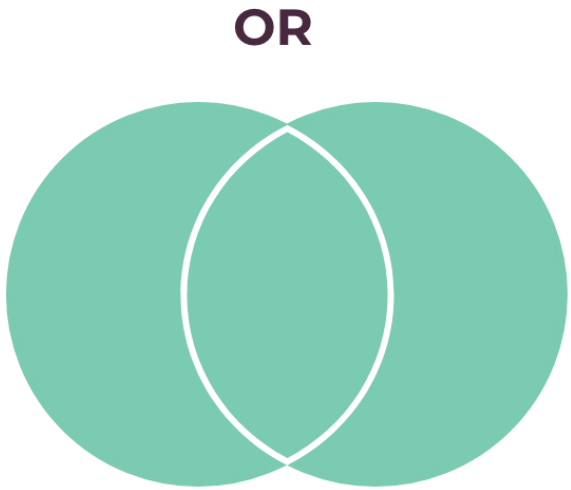
Literature Search

How to use Boolean Operators

2. Using "OR"

- Purpose: To *broaden* your search by including sources that contain at least one of the terms.
- Search Query: "wayfinding" OR "spatial navigation" OR "environmental graphics"
- Explanation: This search will return results that include any of the terms "wayfinding", "spatial navigation", or "environmental graphics."

→ Essential when reviewing literature across different disciplines, where researchers often use different terms for the same spatial phenomenon.



https://recruitinginnovation.com/understanding-boolean-search-and-boolean-operators/

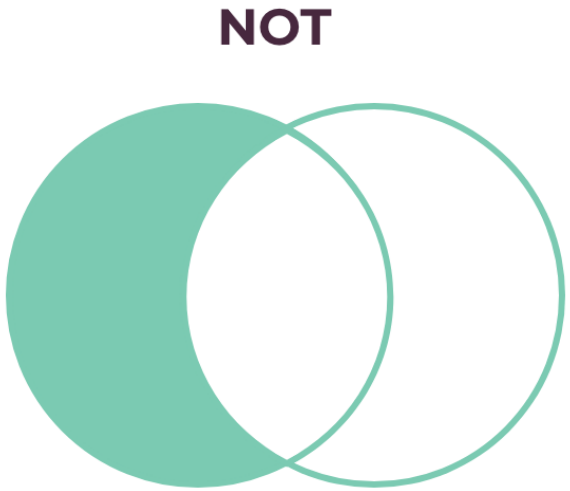
Literature Search

How to use Boolean Operators

3. Using “NOT”

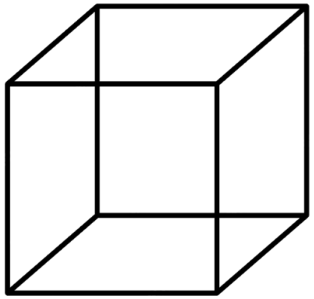
- Purpose: To exclude certain terms from your search results.
- Search Query: “social media” AND “mental health” NOT “advertising”
- Explanation: This search will return results that include "social media” and “mental health” but will exclude any results that mention “advertising.”

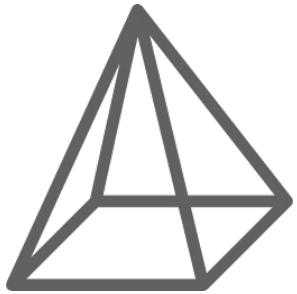
→ Useful when focusing on the psychological aspect of social media while excluding those related to advertising content.



https://recruitinginnovation.com/understanding-boolean-search-and-boolean-operators/

Research Core





By identifying the relative aspects that can help determine the area of research, the ‘scope’ of research can be articulated.

Literature / Background review

‘**Justification**’ is found;
‘**Gap**’ is identified

Questions can be developed

Designing Research

- Once the background/literature review is done, a research gap is found, the justification for the research is articulated, and the research question is ready, the research design and feasibility need to be planned.
- Based on the research question, determine the study design, including the methodology, data types, study participants, and analysis approach, along with the timeline and ethics compliance.



https://ideascale.com/blog/what-is-research-design/

Types of Research – Info Sources

PRIMARY RESEARCH

- Direct & Hands-On
- Gathering first-hand evidence to answer specific, project-bound questions and uncover fresh user insights.
- Collecting brand-new, original data directly from the source.

SECONDARY RESEARCH

- Context & Strategy
- Building essential background knowledge, evaluating existing visual strategies in similar contexts, and grounding in established, field-tested evidence.
- Analyzing, interpreting, and summarizing research and data that others have already gathered.

TERTIARY RESEARCH

- Overviews & Indexing
- Organizing and synthesizing primary and secondary sources to provide a high-level summary.
- Getting a quick overview of a topic, mapping out broad cultural or project contexts, and locating specific primary and secondary sources to investigate more deeply.

How: Research Methodologies

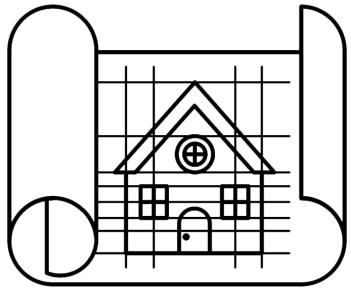
METHOD

- A way of acting or performing tasks, particularly in a systematic or routine way—an organized approach toward achieving a goal.

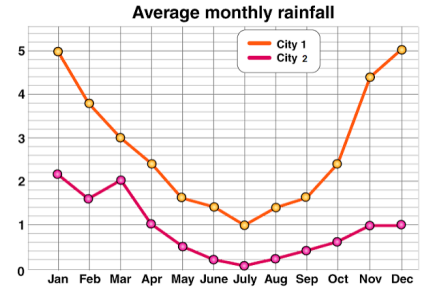


METHODOLOGY

- A body of methods employed in a particular activity
- A structured, predefined approach for executing a communication design project



Types of Research - Data

Qualitative Research	Gathers <u>words, opinions, and experiences</u> through interviews or chats to understand thoughts and reasons.	It is appropriate for gathering deep personal feelings, opinions, and experiences instead of statistics. 
Quantitative Research	Gathers <u>numbers and measurements</u> using surveys, tests, or math to track trends and test ideas.	It relies entirely on hard numbers, percentages, and math to find patterns. 

Ethical Consideration in Research

- Before start taking actions to conduct research, we should consider whether what we are about to do is ethical.
    - ✓ The subject matter of the study
    - ✓ The research methods and procedures



Ethical Consideration in Research

- An ethical framework is established in research to avoid:

    1. A lack of concern regarding the welfare of the research participants;
    2. Misinforming the stakeholders about the nature of the research (that could translate into embarrassing or painful experiences);
    3. Making the participants feel obliged to participate on the research;
    4. Violating the participants, creating deception, and providing falsification of the results.



Ethical Consideration in Research

We need to:

[When planning research]

- ✓ Identify a research problem that will benefit society participants
- ✓ Disclose purpose of the study
- ✓ Not pressure participants into signing consent forms
- ✓ Respect norms and characters of indigenous societies
- ✓ Be sensitive to needs of vulnerable populations (e.g., children)



Ethical Consideration in Research

We need to:

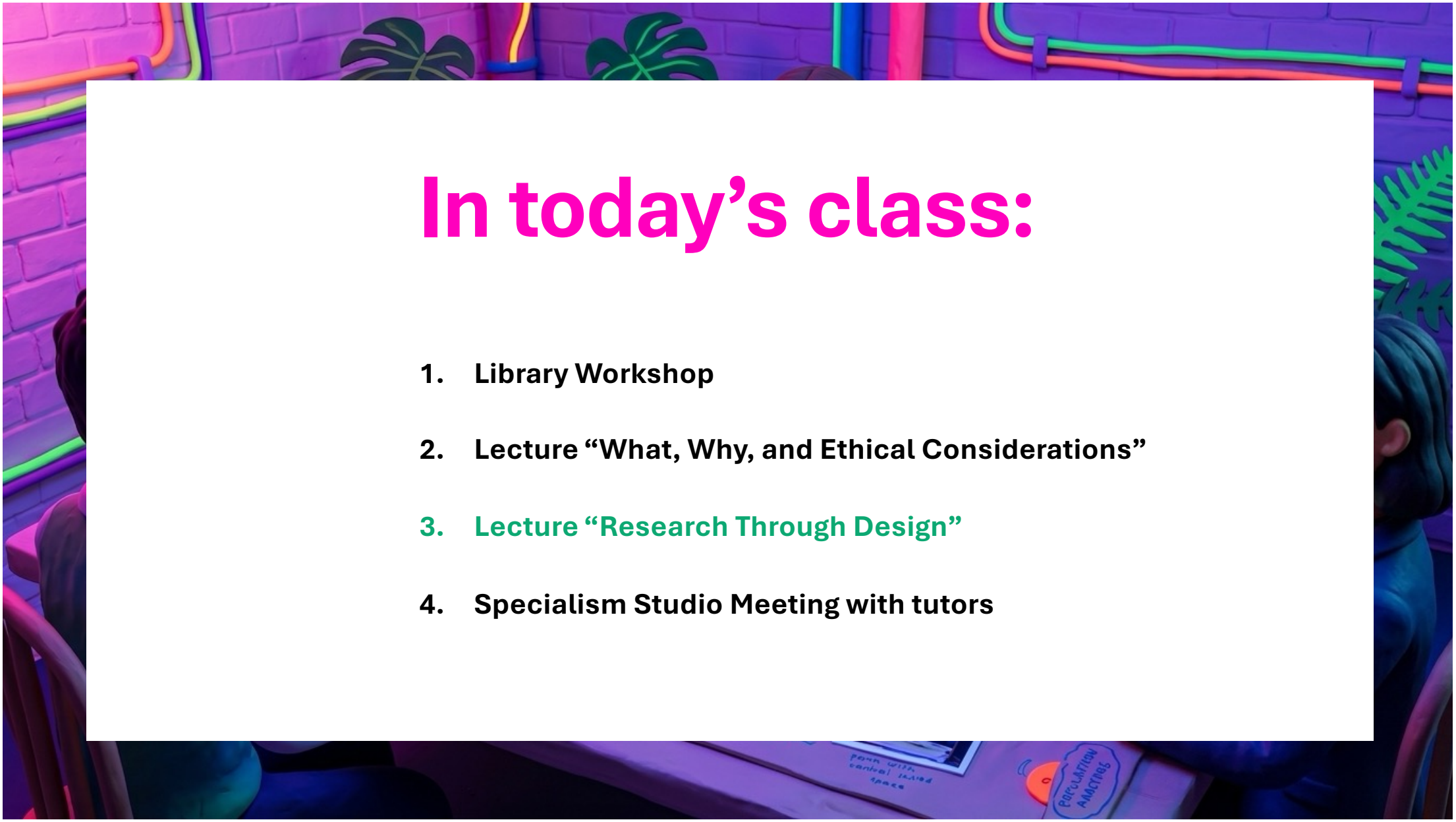
[When collecting data]

- ✓ Respect the site, and disrupt as little as possible
- ✓ Make certain that all participants receive the same treatment
- ✓ Avoid deceiving participants
- ✓ Respect potential power imbalances and exploitation of participants (e.g., interviewing, observing)
- ✓ Do not “use” participants by gathering data and leaving site
- ✓ Avoid collecting harmful information



Types of Design Research - Relationship

Research *into* Design	• *Observes the design world from a distance to understand its history, behavior, and cultural impact*. • Analyzes the context, culture, sociology, or aesthetics of design from an external perspective. • Results typically in a traditional written thesis or dissertation.	Designed artifact is Deconstructed / Critiqued
Research *for* Design	• Involves gathering reference materials, user needs, and technical data to *inform a future project*. • Operates as background preparation - such as market research, ergonomics, or material specs - where the final output is the design artifact itself.	Designed artifact is the intended end goal
Research *through* Design	• *Uses the actual process of making and designing as a primary method of inquiry*. • Includes material exploration, digital development work, and action research. • Combines studio-based experimentation with a written report to document tacit knowledge.	Designed artifact is an active testing tool

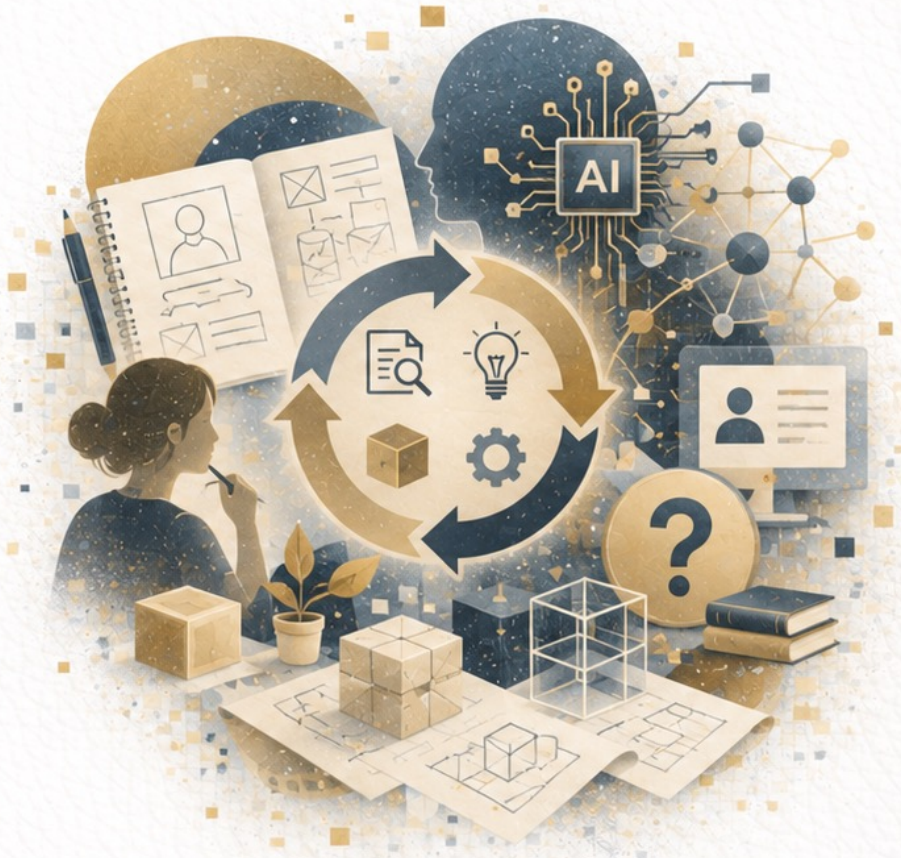


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Using Research Through Design (RtD) Methodology to Address Research Questions Regarding Intelligent System Design

Dr. Rui He



- Who proposed the concept of Research-through-Design (RtD)?
- What is Research-through-Design (RtD) and How it works?
- When can we use Research-through-Design Work?
- Demonstrating Research-through-Design Framework with My PhD Study.

Who proposed the concept of Research-through-Design (RtD).

Research through Design as A New Design Research Methodology Proposed by John Zimmerman in CHI 2007



Research Through Design as a Method for Interaction Design Research in HCI

John Zimmerman, Jodi Forlizzi, Shelley Evenson
Human-Computer Interaction Institute and The School of Design
Carnegie Mellon University
{johnz, forlizzi, evenson}@andrew.cmu.edu

ABSTRACT
For years the HCI community has struggled to integrate design in research and practice. While design has gained a strong foothold in practice, it has had much less impact on the HCI research community. In this paper we propose a new model for interaction design research within HCI. Following a research through design approach, designers produce novel integrations of HCI research in an attempt to make the *right* thing: a product that transforms the world from its current state to a preferred state. This model allows interaction designers to make research contributions based on their strength in addressing under-constrained problems. To formalize this model, we provide a set of four lenses for evaluating the research contribution and a set of three examples to illustrate the benefits of this type of research.

Author Keywords
design, interaction design, interaction design research, HCI research, research through design, wicked problems, design theory, design method

ACM Classification Keywords
H5.2. User Interfaces: Theory and methods.

INTRODUCTION
In recent years we have both witnessed and participated in the struggle as several academic institutions have attempted to integrate design, with technology and behavioral science in support of HCI education and research. While there has been great excitement about the benefits integrating design can bring, we quickly realized that no agreed upon research model existed for interaction designers to make research contributions other than the development and evaluation of new design methods. Over the last two years we have undertaken a research project to (i) understand the nature of the relationship between interaction design and the HCI research community, and (ii) to discover and invent methods for interaction design researchers to more effectively participate in HCI research.

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CHI 2007, April 28–May 3, 2007, San Jose, California, USA.
Copyright 2007 ACM 978-1-59593-593-9/07/0004...$5.00.

Through our inquiry we learned that many HCI researchers commonly view design as providing surface structure or decoration. In addition, we lack a unified vision of what design researchers can contribute to HCI research. This lack of a vision for interaction design research represents a lost opportunity for the HCI research community to benefit from the added perspective of design thinking in a collaborative research environment. The research community has much to gain from an added design perspective that takes a holistic approach to addressing under-constrained problems.

To address this situation, this paper makes two contributions: (i) a model of interaction design research designed to benefit the HCI research and practice communities, and (ii) a set of criteria for evaluating the quality of an interaction design research contribution. The model is based on Frayling's *research through design* [14], and it stresses how interaction designers can engage "wicked problems" [21]. What is unique to this approach to interaction design research is that it stresses design artifacts as outcomes that can transform the world from its current state to a preferred state. The artifacts produced in this type of research become design exemplars, providing an appropriate conduit for research findings to easily transfer to the HCI research and practice communities. While we in no way intend for this to be the only type of research contribution interaction designers can make, we view it as an important contribution in that it allows designers to employ their strongest skills in making a research contribution and in that it fits well within the current collaborative and interdisciplinary structure of HCI research.

Definitions
As we conducted this inquiry, we quickly realized that within both the HCI and design communities there is an inconsistent and confusing use of the following terms. Therefore, below we provide a set of definitions for these terms with respect to this paper.

Designer. Using such a generic term is a challenge at best. At CHI 2006's SIG: "The CHI Design Community", Bill Buxton sarcastically claimed that if everyone is a designer because they select their own clothes, then everyone is also a mathematician, because we all count our change. His comment captures what a loaded term "designer" is. Within



Prof. John Zimmerman

Tang Family Professor of Artificial Intelligence and Human-Computer Interaction

Human-Computer Interaction Institute School of Design

Carnegie Mellon University

What is Research-through-Design (RtD) and How it works?

RtD emphasizes the creation of artifacts that "*transform the world from its current state to a preferred state*" (Zimmerman et al., 2007) . These artifacts, designed for research purposes, serve as conduits for reflection and evaluation, with the goal to "*produce knowledge for the research and practice communities*" (Zimmerman et al., 2007).

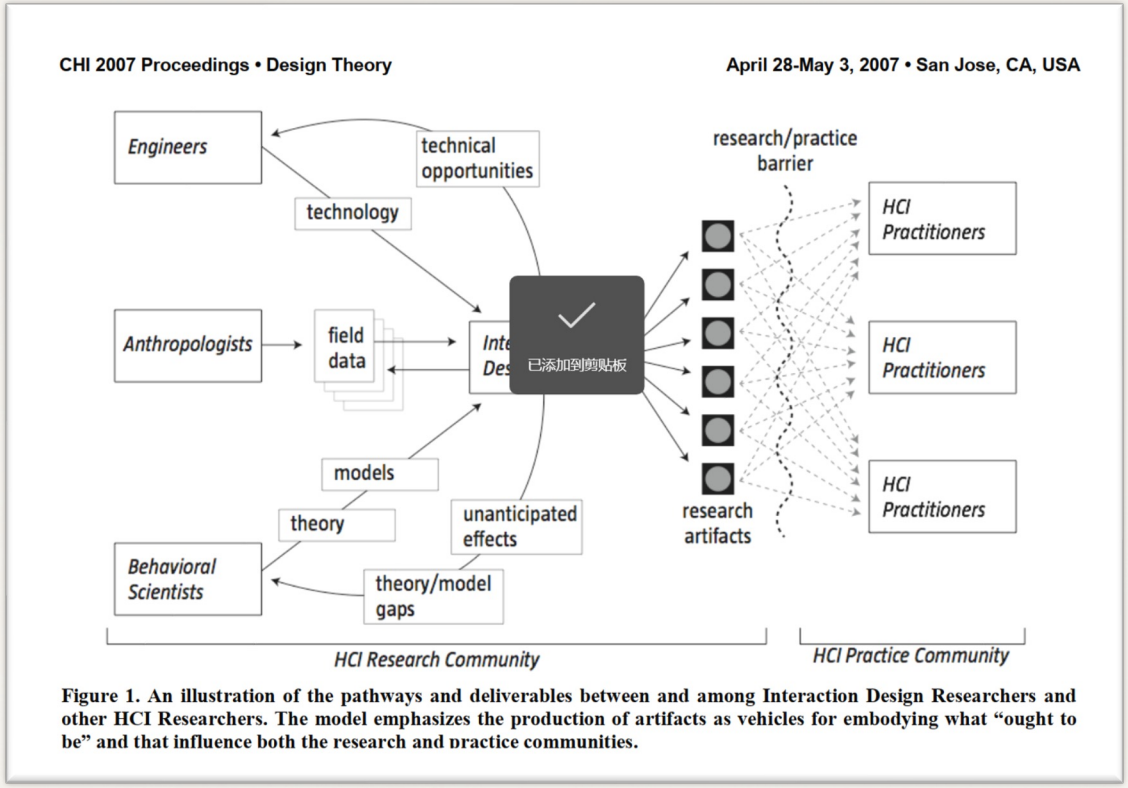
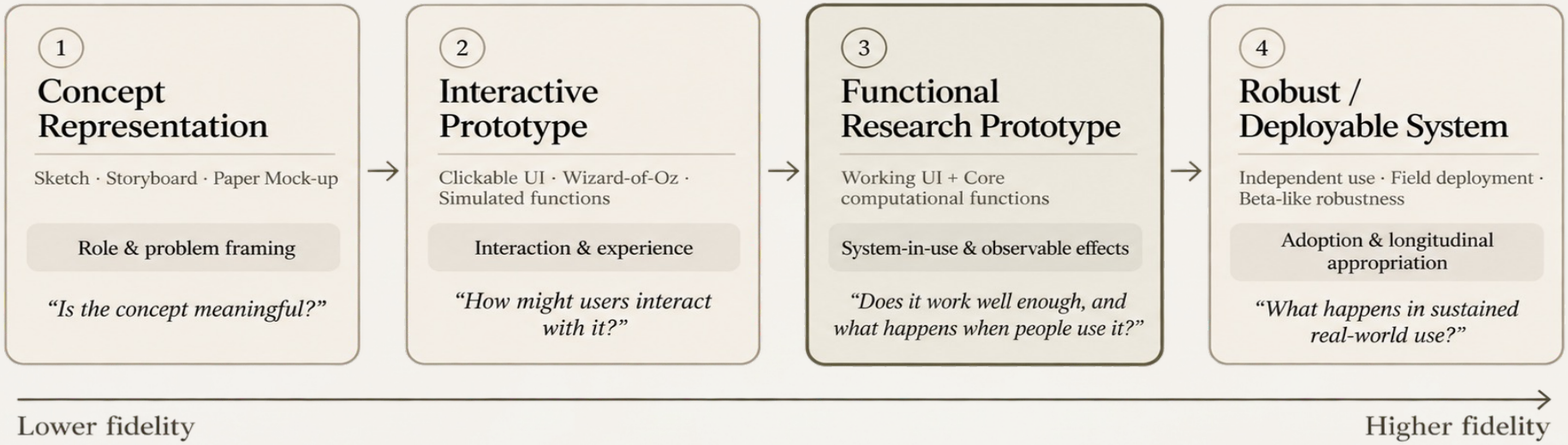


Figure 1. An illustration of the pathways and deliverables between and among Interaction Design Researchers and other HCI Researchers. The model emphasizes the production of artifacts as vehicles for embodying what "ought to be" and that influence both the research and practice communities.

(Zimmerman et al., 2007, p. 498)

Phase 1 — Design Exploration and Artifact Development

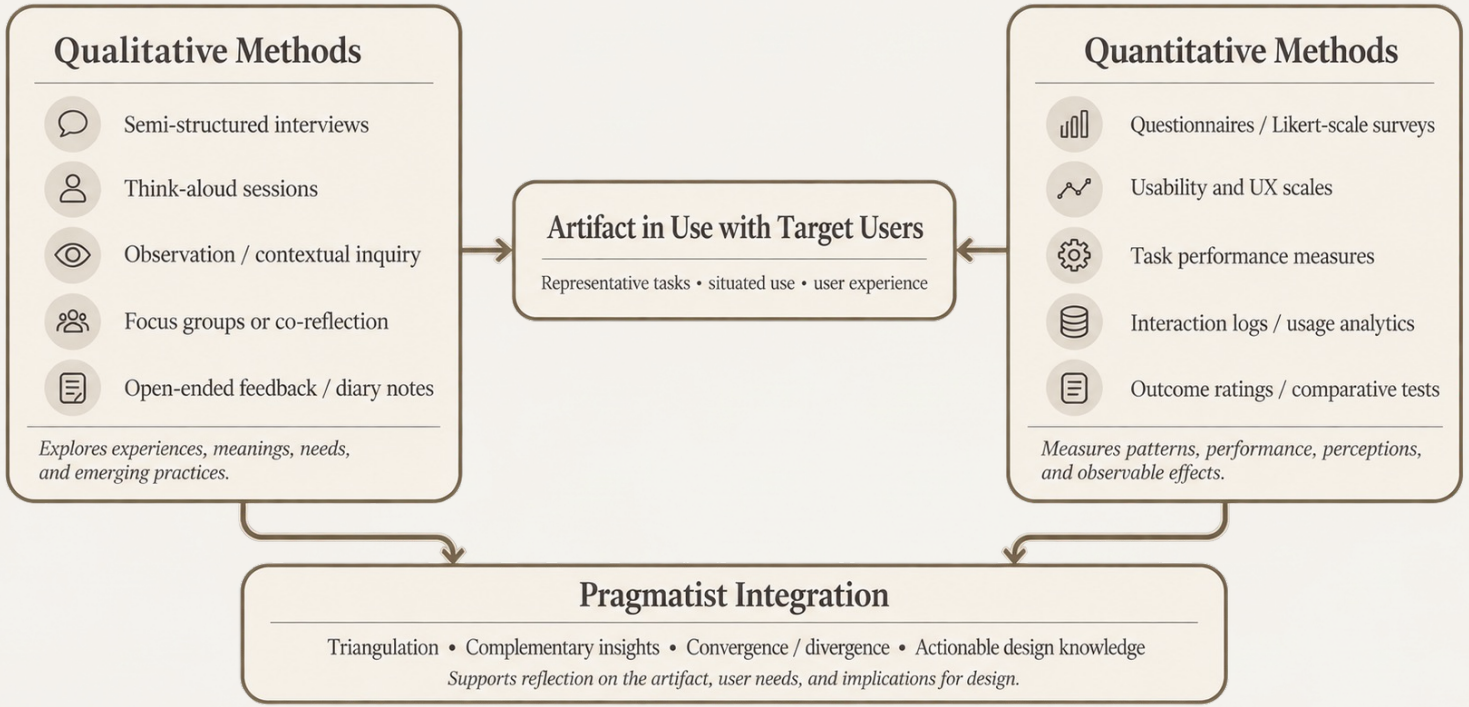
This phase concerns the framing and exploration of the design problem and the iterative development of a research artifact. Drawing on relevant theoretical knowledge, contextual understanding, and technological opportunities, the researcher engages in cycles of ideation, prototyping, critique, and reframing to materialize a proposed preferred state.



How much should we build? --- to the level necessary to answer the research question. (Hudson & Mankoff, 2014)

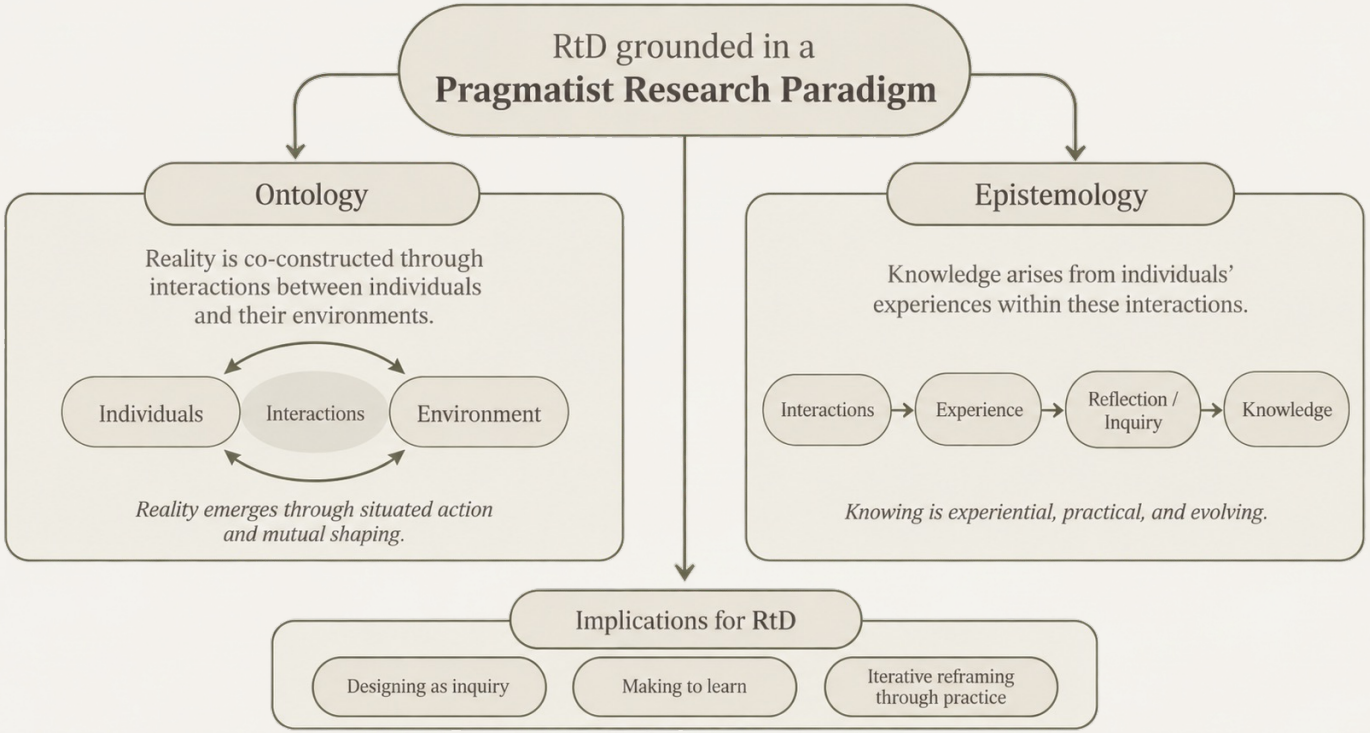
Phase 2 — Artifact-mediated Inquiry and Knowledge Generation

This phase concerns inquiry through the situated use of the research artifact. By examining how people interact with and experience the artifact, the researcher investigates its effects, identifies emergent or unanticipated phenomena, and reflects on what these observations reveal about the design problem, users, and the broader design space. The resulting insights are articulated as transferable design knowledge, implications, or theoretical propositions. (Hudson & Mankoff, 2014)



When can we use Research-through-Design Work?

Research-through-Design is a methodology framework for addressing research questions with a pragmatism paradigm.

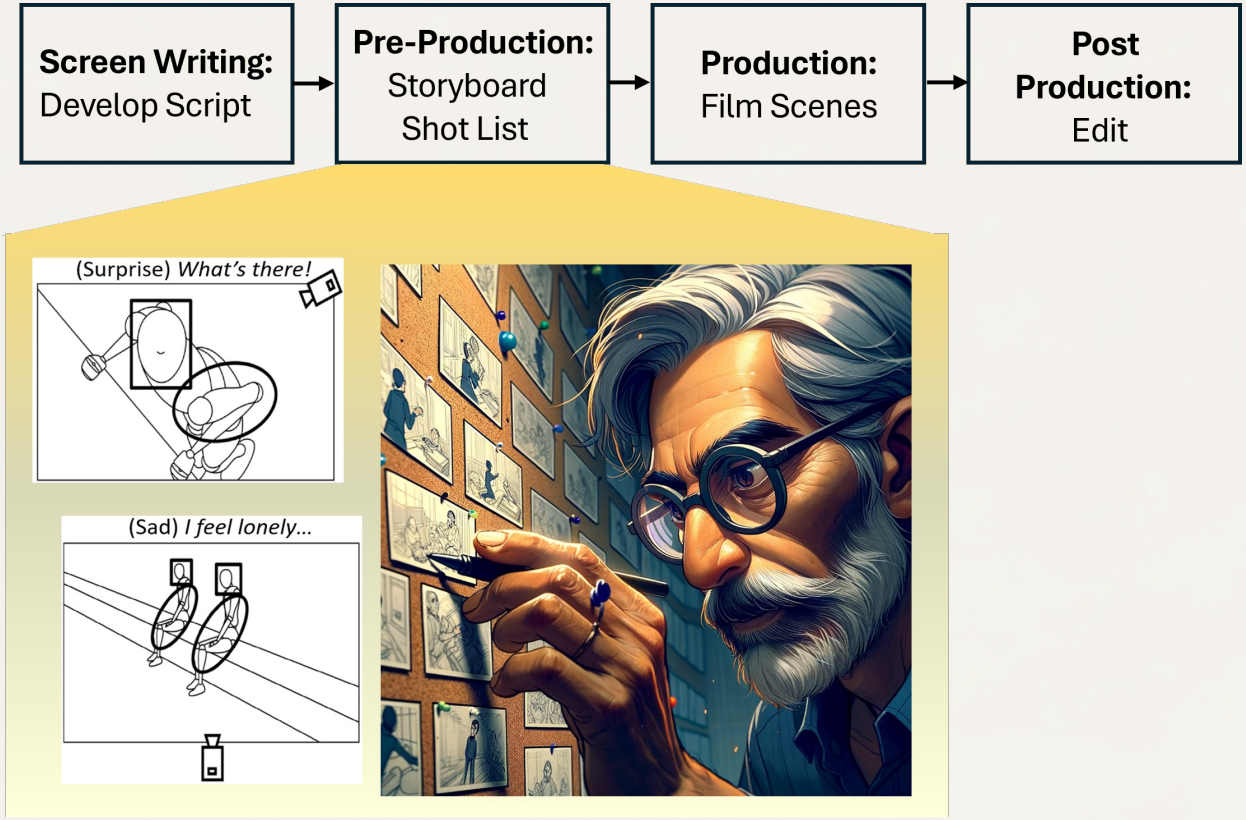


The nature of research questions concerning ***design practice*** and ***design thinking*** therefore aligns with the spirit of pragmatism.

Demonstrating Research-through-Design Framework with My PhD Study:

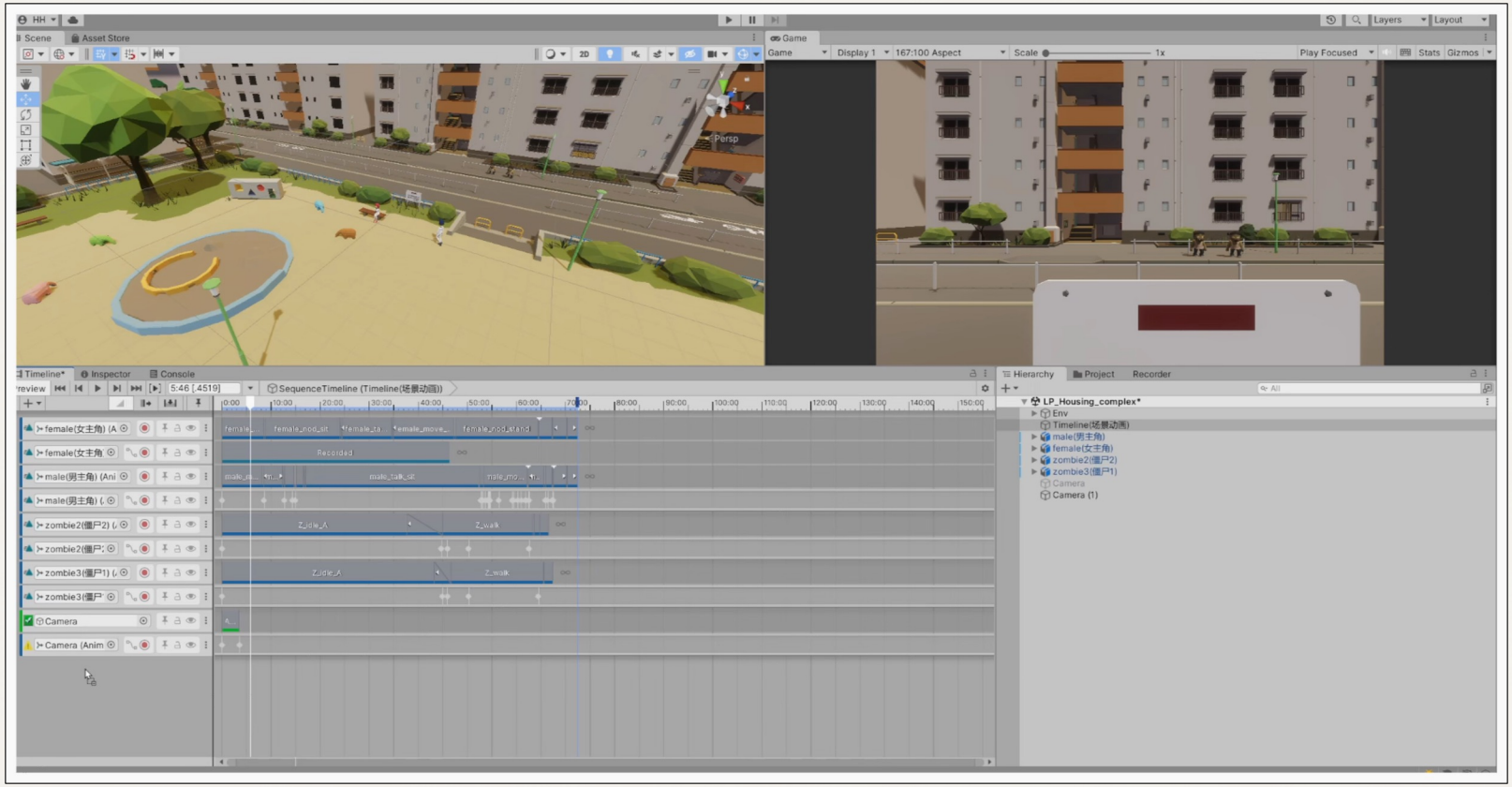
Supporting Creative Design for 3D Cinematic Composition with Generative Artificial Intelligence

Traditional Filmmaking: Cinematic Composition via Storyboard



Game-engine-based Filmmaking:

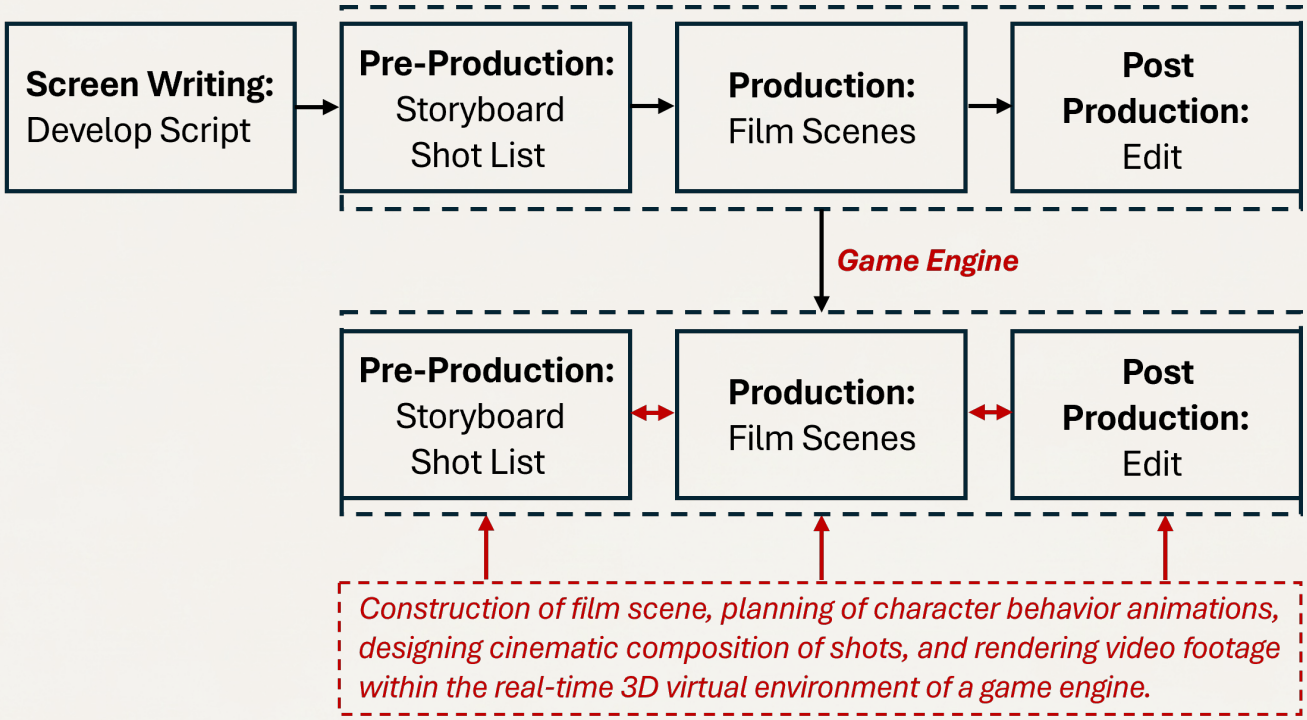
Initially Adopted for Producing Digital Game Cutscenes



Explorative Cinematic Composition Design & Evaluation in Real-Time Virtual Environment

Game-engine-based Filmmaking:

Workflow Compared with That of Traditional Filmmaking



Game-engine-based Filmmaking:

Widely Employed in the Pre-Production Stage of Traditional Filmmaking



1. Look for potential cinematic composition design alternatives on digital filmmaking platform using prototype 3D animation



2. Confirm the final shots design on storyboard



3. Produce the formal shot with formal artwork

"Literally see and try-out rather than imagining to achieve creative outcomes…"

Anno, H. (Evangelion: 3.0+1.01 Thrice Upon a Time, 2021), 22:52



Game-engine-based Filmmaking:

A Thriving Method for Producing AIGC Films Guided by Fine-tuned Cinematic Composition



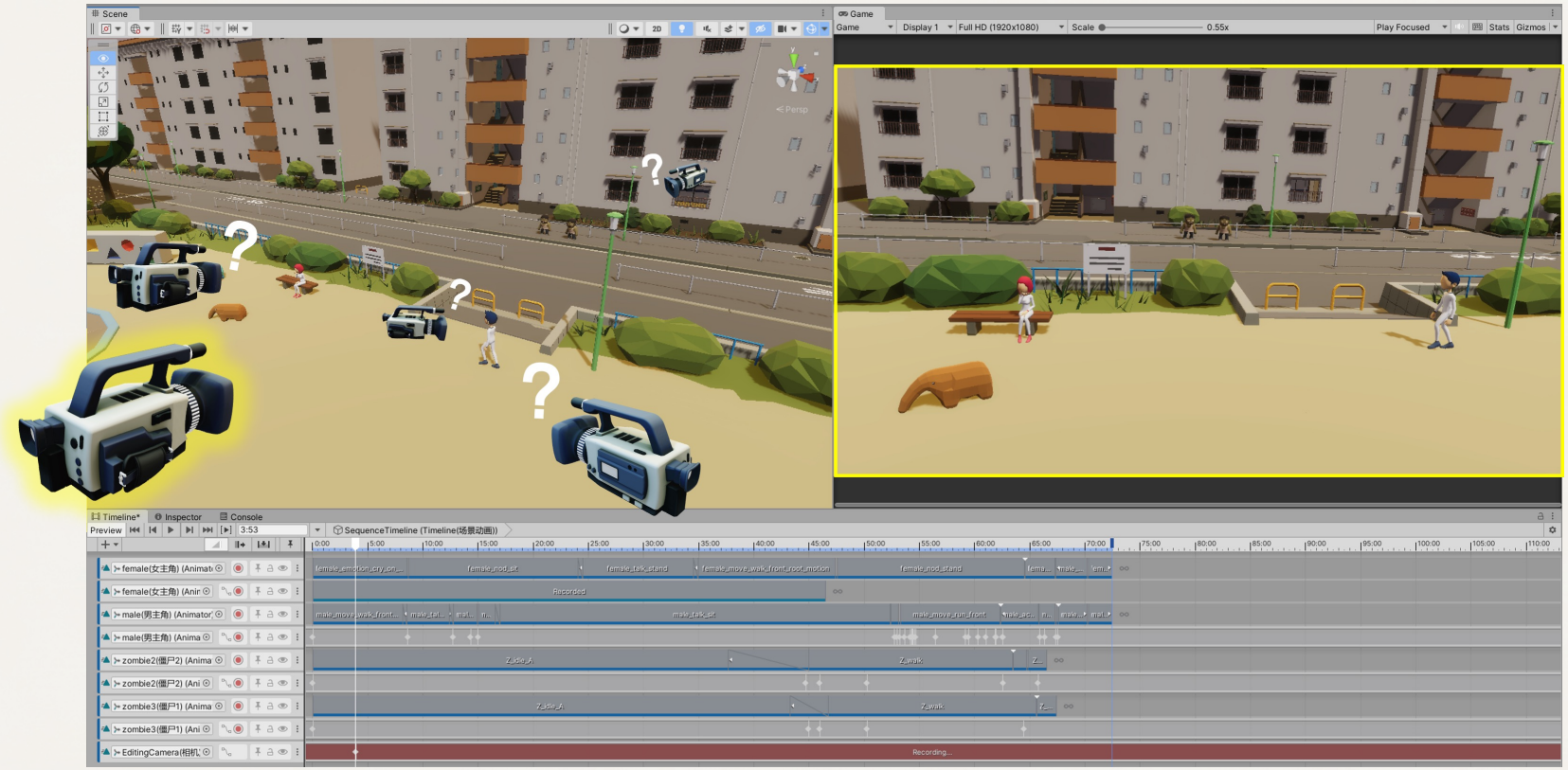
Prototyping a film scene and the cinematic compositions in game engine using interim virtual characters



The screenshot/video rendered in game engine serves as visual prompts that guide the production of AIGC film shots

Game-engine-based Filmmaking:

Creative Cinematic Composition Remains Challenging…

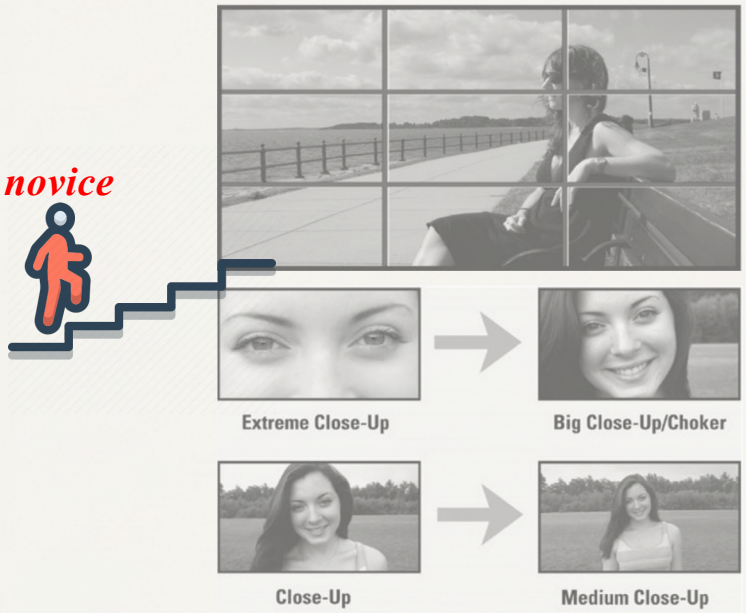


Trying-out of different design alternatives explicitly in a *situated environment*

Game-engine-based Filmmaking:

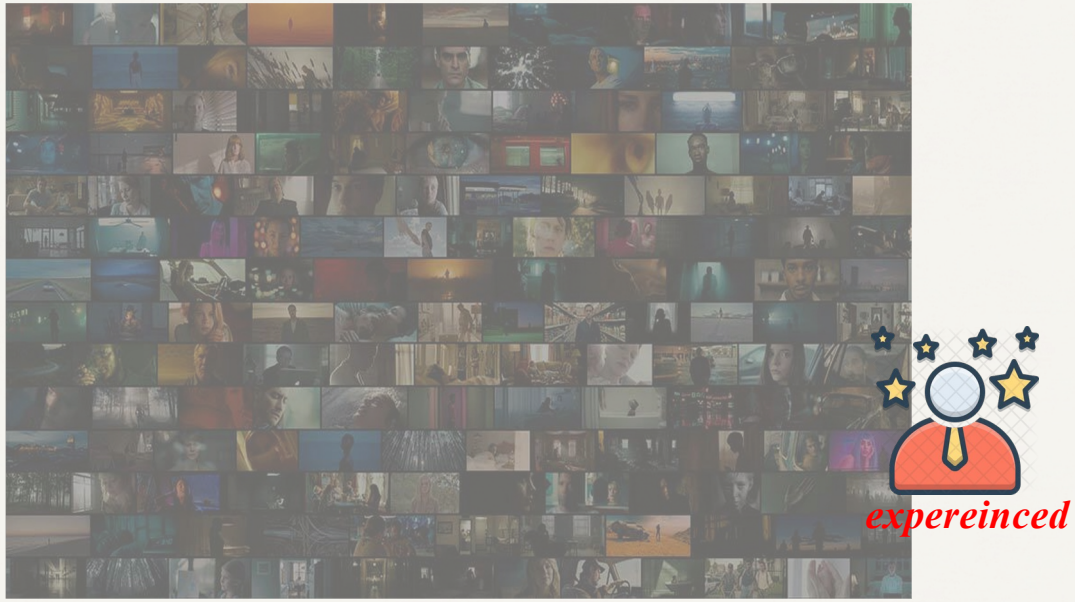
Creative Cinematic Composition Remains Challenging...

Follow standard *rules* (Bowen, 2013)



canonical design: less creative...

Lean from *examples* by film analysis (Aumont et al., 1992; Sobchack and Sobchack, 1987)



stylized cinematic variation: more creative but the learning process remains subjective, lengthy and challenging...

Game-engine-based Filmmaking:

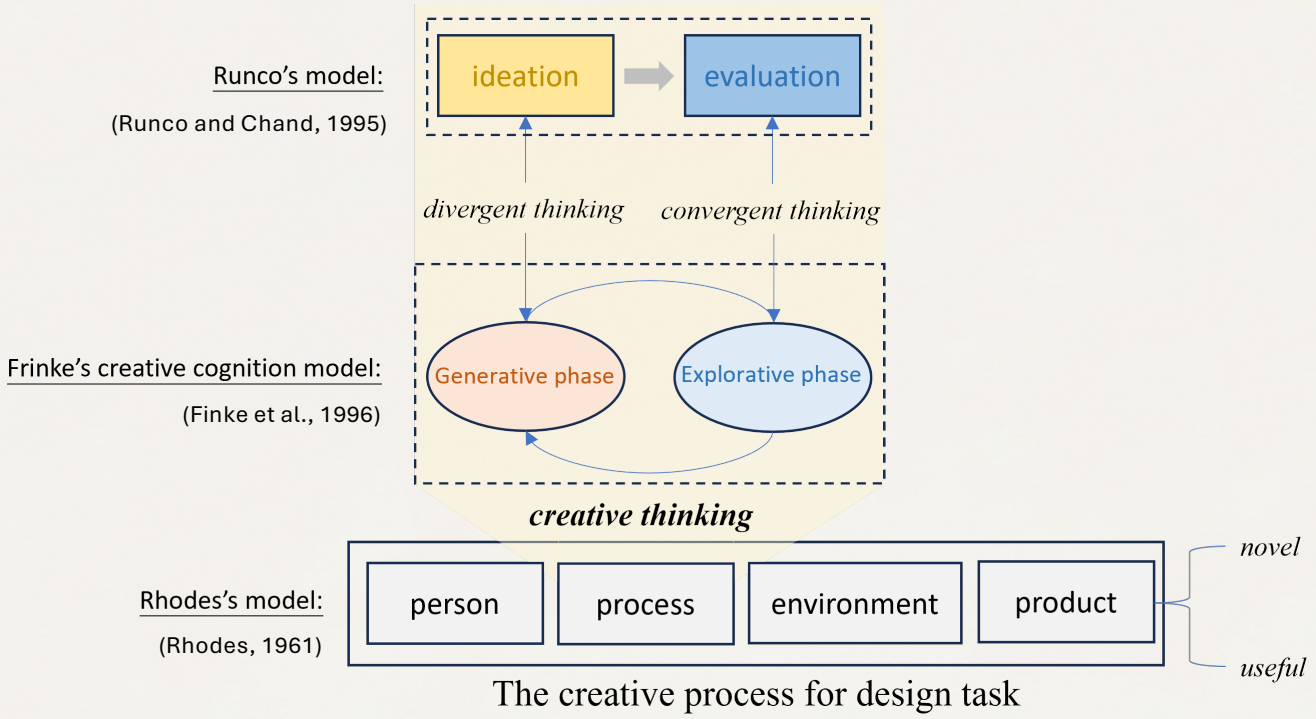
Creative Cinematic Composition Remains Challenging...

A *situated* *Creativity* is entailed for cinematic shot composition design within the 3D real-time environment (Davis et al., 2011).

Presentation Overview

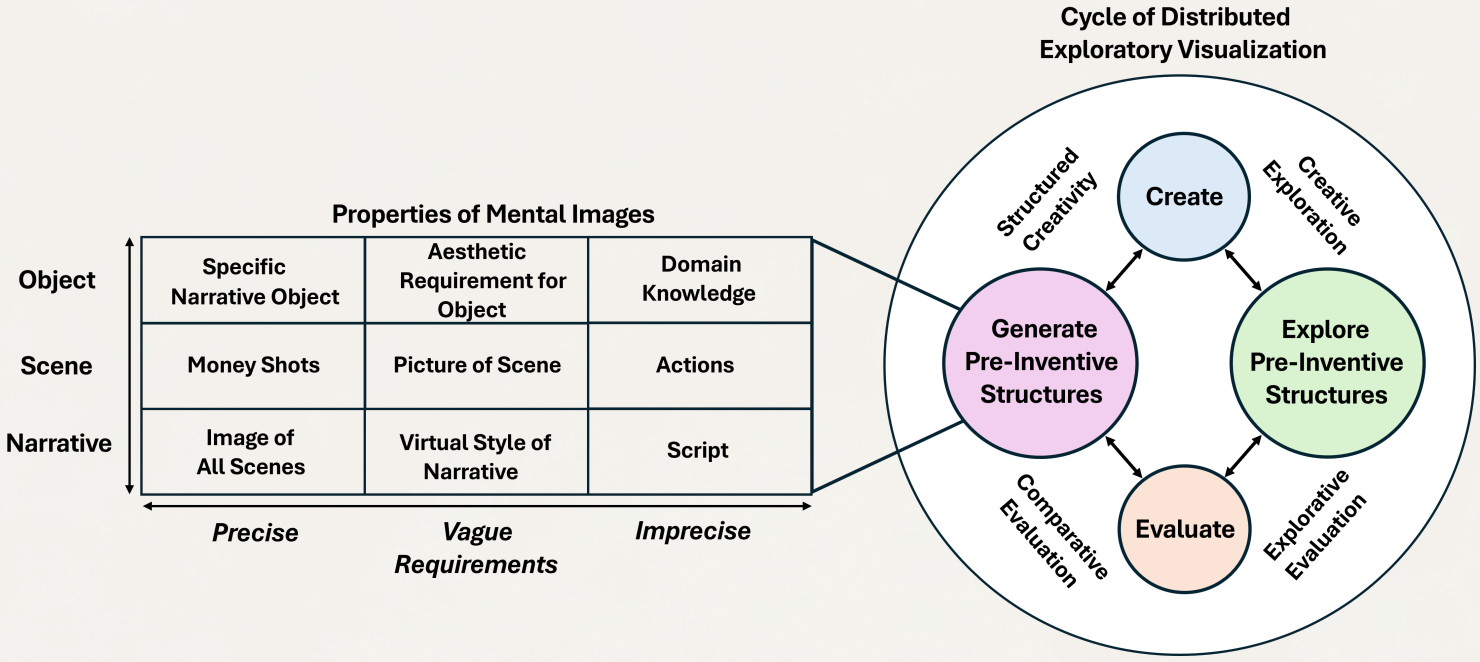
- Background
- **Theories, Related Studies, and Research Questions**
- Methodology, Research Procedures, and Results
- Findings, Limitations, and Future Work

Modern Theories of Creativity: Modeling the Mechanisms of Creative Thinking in Design Tasks



The creative process for design task

Davis et al.'s *Distributed Exploratory Visualization Model* Characterizing the Situated Creative thinking Workflow in Game-engine-based Filmmaking



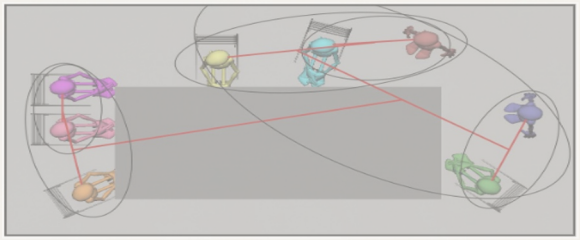
(Davis et al., 2011)

Addressing the Creative Challenge of Cinematic Composition Design in Game-engine-based Filmmaking: Related Work and Methods

Automatic Systems Based on Heuristic Rules and Data-Driven Artificial Intelligence



(Jiang et al., 2020)

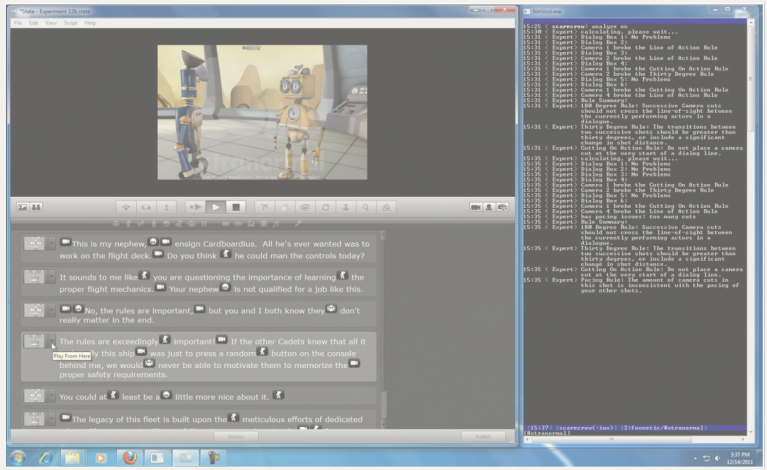


(Kardan and Casanova, 2008)

⋮

Lacking interactivity mechanisms through which human user can inject creative thinking...

Creativity Support Tools (A Hot Spot of HCI-oriented Creativity Research & A Branch of Human-AI Co-Creative System)



(Davis et al., 2013)

Focusing on guiding beginner filmmakers to produce correct work but lacks support for experienced filmmakers in exploring and evaluating creative, novel ideas...

Nowadays Popular AIGC Models



canonical design: less creative...

General-purpose generative models like Stable Diffusion and Sora lack the spatial precision, direct control, and real-time interactivity needed for camera poses in a 3D environment...

Research Questions of My PhD study

RQ1: How to design and implement a computational system functioning as a creativity support tool to address the design challenge for producing creative cinematic compositions in game-engine-based filmmaking?

RQ2: How does the adoption of such a system impact the creative thinking processes and design outcomes of game-engine-based filmmakers?

RQ3: What can be the latent user needs and preferences in human-AI collaboration within the creative design processes?

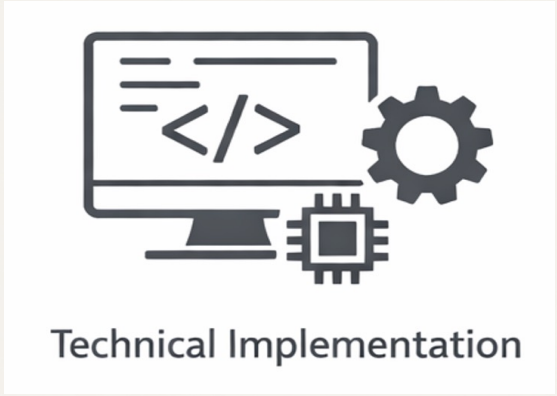
A Pragmatic Human-Computer-Interaction Framework for Studying Creative Practice

When the research question requires designing the research tool itself



Research-through-Design

Design an AI-based creativity support system based on initial needs derived from the creative partitioners.



Technical Implementation

Implement the designed system with feasible computational and AI methods.



Mixed Methods Evaluation

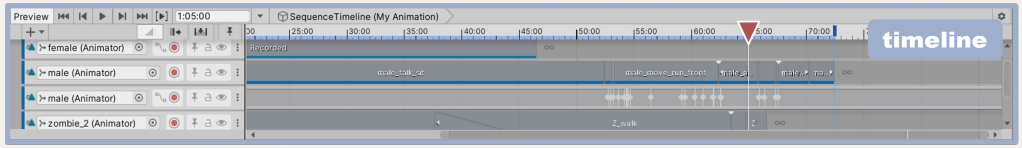
Examine the creative practitioners' creative thinking process and outcomes supported by the system using both qualitative and quantitative methods.

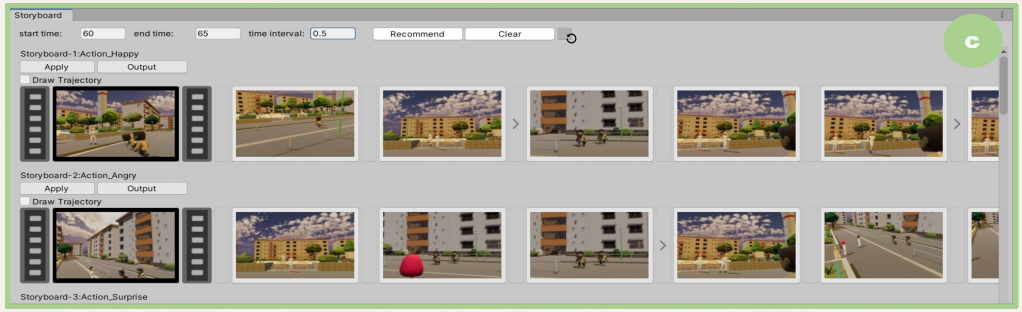
Designing a Creativity Support System (*Cinemassist*) for Cinematic Composition

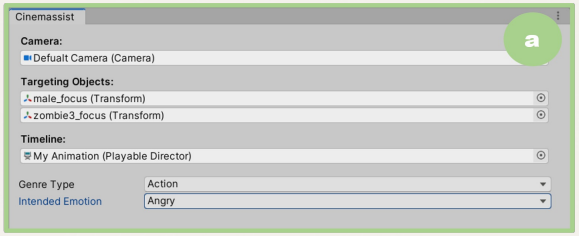
Informed by formative study with professional creative practitioners of digital filmmaking, followed with several iterations of quick prototyping, ensuring the interaction design decisions are grounded in implementable system behaviors and closely aligned with the technical constraints













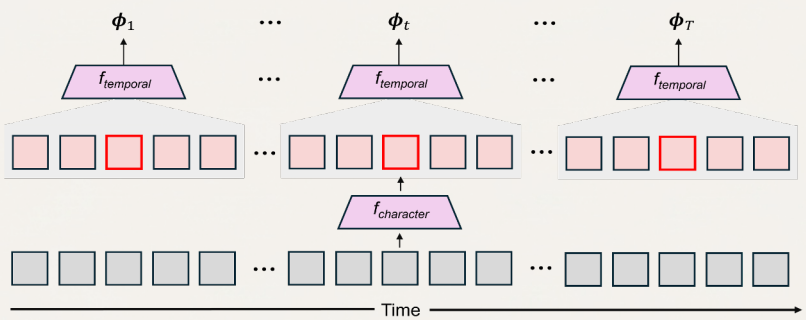
*The **storyboard panel**, which supports cinematic composition design across a selected time span by suggesting alternative sequences of visual compositions, enabling users to compare and reflect on different creative directions.*

*The **control panel**, which allows users to load 3D animations and set overall creative goals, such as the type of story and the intended emotional tone.*

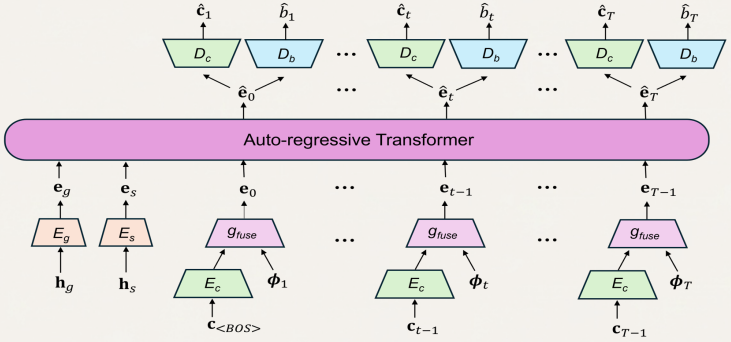
*The **design panel**, which supports moment-by-moment cinematic composition by allowing users to capture, refine, and compare different visual framing options during the 3D animation.*

Technical Implementation of the Creativity Support System

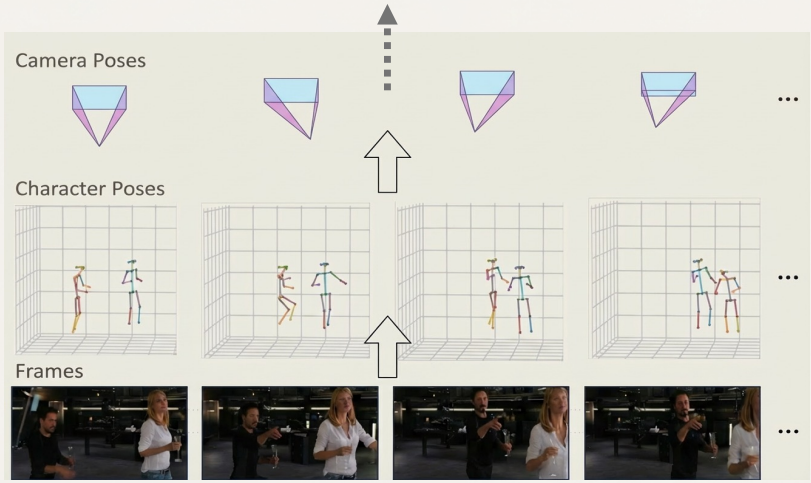
A customized machine learning architecture trained on cinematic data to support real-time creative decision-making



(a) Keyframe feature extractor



(b) Camera Pose Generator

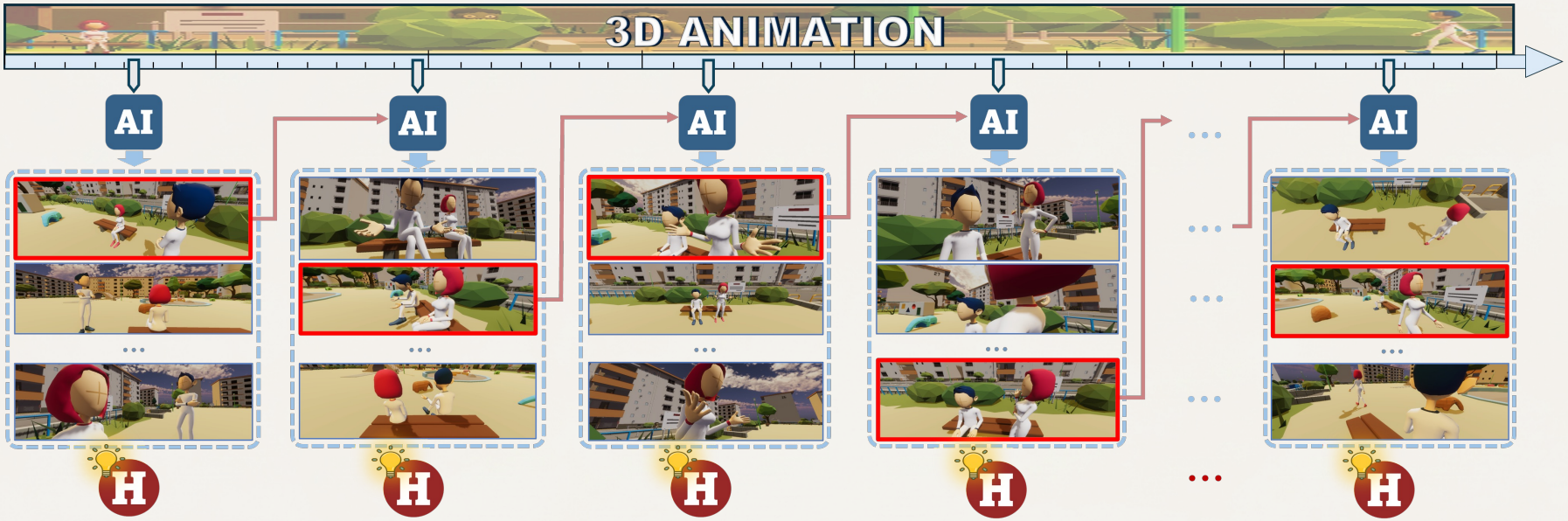


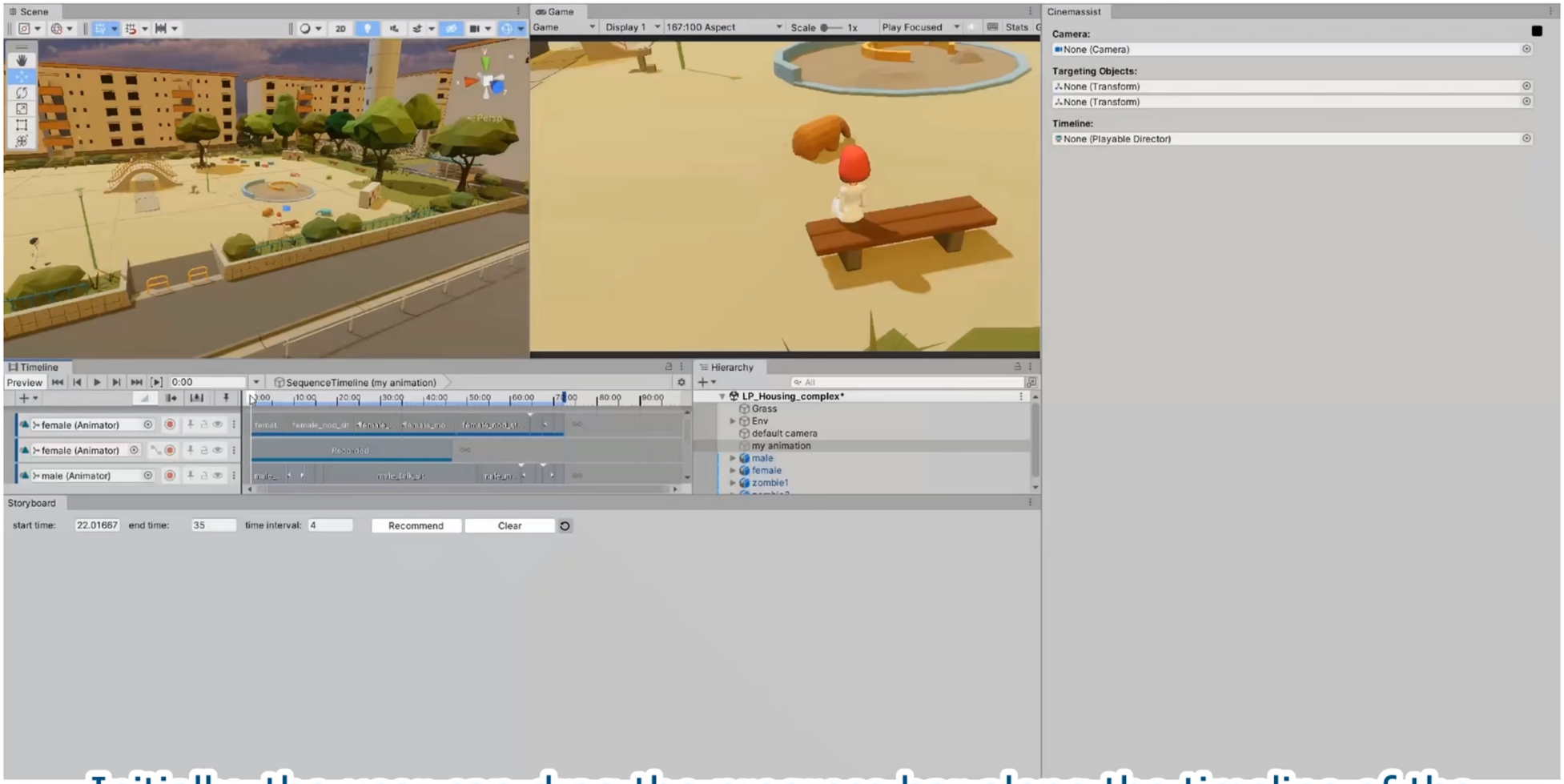
Extracting the 3D camera poses corresponding to different character animations from the training scenes

Movie Name	Genre	Valid Samples
Titanic	Romance	51
Juno	Romance	54
Twilight	Romance	63
Pretty Woman	Romance	72
English Patient	Romance	49
Pride & Prejudice	Romance	47
Seven	Thriller	39
Mulholland Drive	Thriller	48
The Usual Suspects	Thriller	32
Blade Runner	Thriller	20
The Butterfly Effect	Thriller	85
The Silence of the Lambs	Thriller	31
Avatar	Action	28
Inception	Action	93
The Avengers	Action	72
The Dark Knight Rises	Action	84
Gladiator	Action	58
The Lord of the Rings: The Return of the King	Action	51

A Training Dataset of Trending Films

Workflow Demonstration of the Implemented Creativity Support System





Initially, the user can drag the progress bar along the timeline of the

Mixed Methods Evaluation

A mixed-methods evaluation combining controlled tasks, self-reported confidence, debrief interview and expert assessment.

Participants

18 participants

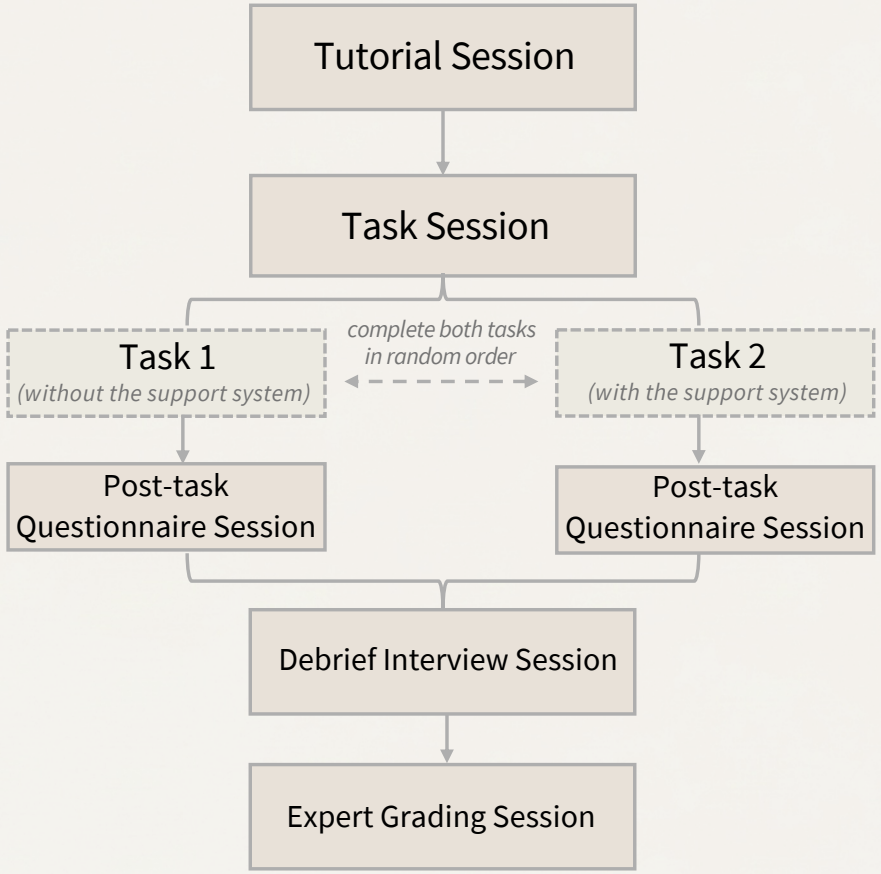
- Animation-related background (n = 9)
- Non-animation background (n = 9)
- Experience with Unity / Unreal / 3D tools

Experts

2 professional filmmaking experts

- Independent grading of the cinematic results produced by the 18 participants

Study Procedures

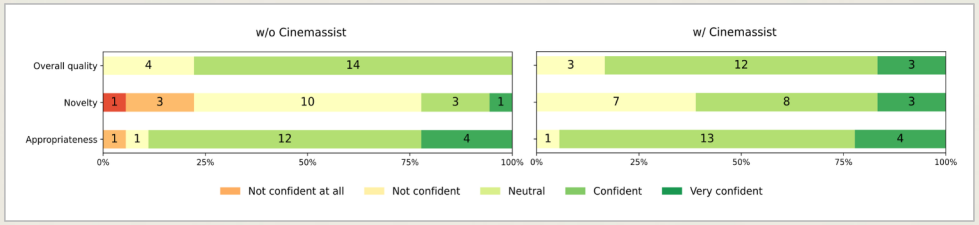


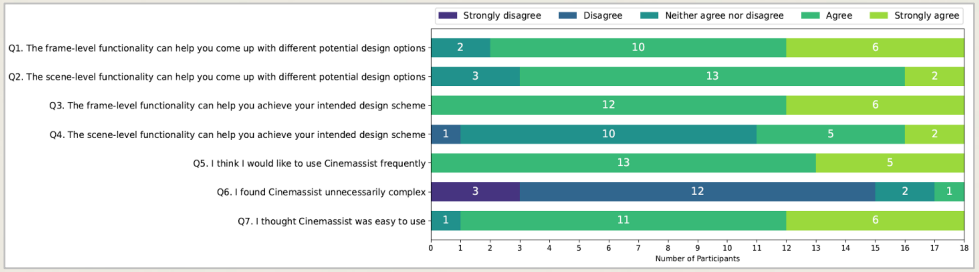
Data Collection and Data Analysis Approaches

	Collected Data	Data Type	Analytic Purpose	Analysis Approach
Task Session	Outcome of task sessions	Qualitative	Evaluate creative outcomes	Reviewed by experts during grading sessions
	Screen recordings of task sessions	Qualitative	Understand creators' on-screen behavior	Qualitative review to supplement observation notes
	Task completion time	Quantitative	Measure creation efficiency	Descriptive statistical analysis
	Observation notes	Qualitative	Observe creators' behaviors during creation	Inductive thematic analysis of behaviors *and interactions*
Post-task Questionnaire Session	Post-task questionnaire ratings	Quantitative	Measure creators' subjective confidence rating	Descriptive statistical analysis of Likert-scale responses
	Post-task questionnaire comments	Qualitative	Interpret creators' subjective confidence rating rationale	Qualitative annotation of subjective confidence ratings
Debrief Interview Session	Semi-structured *debrief* interviews	Qualitative	Understand creators' reflective thinking about their creative process	Inductive thematic analysis, informed by prior theory
Expert Grading Session	Expert questionnaire ratings	Quantitative	Assess creative outcomes through experts' evaluations	Descriptive table of Dichotomous-scale responses
	Expert rating rationale comments	Qualitative	Interpret experts' assessment rating rationale	Qualitative annotation of expert ratings

Data Analytic Results (Evidence Supporting the Findings)

Post-task questionnaire





Expert assessment and task completion time

Animation Background Group									
Participant ID	P1	P2	P3	P4	P5	P6	P7	P8	P9
E1 preference	HA	H	H	HA	H	HA	HA	HA	H
E2 preference	HA	H	HA	H	H	HA	H	HA	HA
Non-Animation Background Group									
Participant ID	P10	P11	P12	P13	P14	P15	P16	P17	P18
E1 preference	H	H	HA	HA	H	HA	H	HA	H
E2 preference	HA	H	H	H	H	HA	H	H	H

Animation Background Group									
Participant ID	P1	P2	P3	P4	P5	P6	P7	P8	P9
E1 preference	H	H	H	HA	H	HA	HA	HA	H
E2 preference	HA	H	HA	H	H	HA	H	HA	HA
Non-Animation Background Group									
Participant ID	P10	P11	P12	P13	P14	P15	P16	P17	P18
E1 preference	H	H	HA	HA	H	HA	HA	H	H
E2 preference	HA	H	H	H	H	H	H	HA	H

Animation Background Group									
Participant ID	P1	P2	P3	P4	P5	P6	P7	P8	P9
E1 preference	HA	H	H	HA	H	HA	HA	HA	H
E2 preference	HA	H	HA	H	H	HA	H	HA	HA
Non-Animation Background Group									
Participant ID	P10	P11	P12	P13	P14	P15	P16	P17	P18
E1 preference	H	H	HA	H	HA	HA	H	HA	H
E2 preference	HA	H	H	H	H	H	HA	H	H

Animation Background Group									
Participant ID	P1	P2	P3	P4	P5	P6	P7	P8	P9
H completion time	81	58	35	41	60	25	33	41	57
HA completion time	60	26	29	50	83	29	24	39	32
Non-Animation Background Group									
Participant ID	P10	P11	P12	P13	P14	P15	P16	P17	P18
H completion time	22	13	37	75	21	36	22	46	44
HA completion time	23	23	39	53	20	34	30	59	41

Thematic analysis results of the debrief interview

- **Human starting exploration directly from automatic divergent inputs (T7-1).**

When compared to the same exploratory practice in the manual workflow without *Cinemassist*'s assistance, 6 participants revealed that *Cinemassist* had drafted the divergent exploration interim products for them. When taking these results as input, they could instantly shift to refine them using these inputs' spatial adjustment handlers and output the final products of the exploration process. For instance, P12 recalled:

> *" If I only spend time on meaningful adjustments rather than having to set up those several angles first, the whole process can become very fast. "*

P9 added:

> *" Even if I messed up the suggested settings, the system could reset them so that I could try once again. "*

As AI took over the initial divergent inputs production, 4 participants confirmed that they had become less conservative in the explorative practice, as they only needed to resume, but not manually initialize a tryout. P11 represented this sentiment:

> *" Since it's fast, I don't hesitate. I became more selective and explored a wider variety of shots. "*

P11 further elaborated on one of these attempts captured in the task's video recording:

> *" Yes, for example, here, I got some complex shots that I would have been too lazy to attempt in the traditional way. "*

- **Human evaluation shifted from sequential to parallel form (T8-1).**

When compared to the same Evaluate practice in the AI-absent workflow, responses from almost two-thirds (13 in total) of the participants suggest a notable shift in their evaluation process. Previously, evaluations were conducted sequentially: assessing one outcome at a time. However, with the presence of Cinemassist, the process transformed into a parallel approach, where multiple outcomes were evaluated simultaneously. Consequently, they were able to complete more evaluation iterations compared to the AI-absent workflow to see which could best convey their design requirements, which were always vague. P14 described this transformation:

> *"Previously, I spent most of the time on preparing a camera pose to evaluate whether it could work, now I spent this time on evaluating which work could best do the storytelling and emotion expression."*

P11 further emphasized the significance of this transformation for the designer:

> *"I evaluated multiple options and went beyond my initial ideas. And that's what I believe my role as a designer should be—not just execution, but focusing on creative decisions."*

- **Necessity, effectiveness and limitations of encapsulating vague design requirements (T8-2).**

Most participants confirmed that the semantic categories—*Genre Type* and *Intended*

Thematic analysis results of the observation notes

- **Targeted refinement replacing wide-range manual creation and exploration.**

Broad exploration, observed in the 15 participants' workflows (P1, P2, P3, P4, P5, P6, P7, P9, P10, P11, P12, P15, P17, P18) during the task without *Cinemassist*'s support, was observed in only 5 participants (P1, P5, P7, P11, P12) in the task with *Cinemassist*'s support. Most participants focused on refining Cinemassist's frame-level or scene-level suggestions rather than engaging in manual creation or large-scale exploration. This observation matches participants' feedback in their *Create* and *Explore Pre-inventive Structures* practices when the *Cinemassist* system was provided (**T6**, **T7**).

- **Distinct strategy of using the two system functions.**

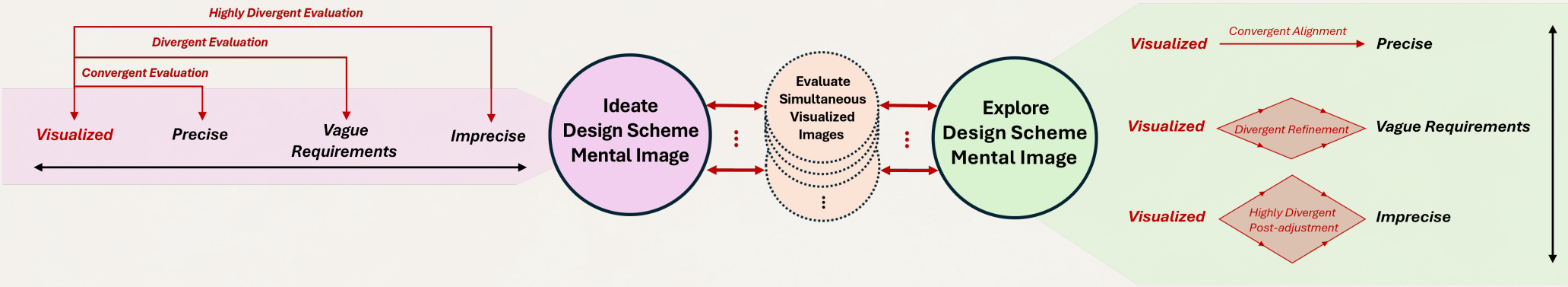
We observed 14 participants (P2, P3, P4, P5, P6, P7, P8, P9, P10, P11, P16, P17, P18) used the scene-level function to generate initial sequences and then refined them with the frame-level function. Conversely, the remaining 4 participants (P1, P12, P13, P15) relied only on the frame-level function and did not use the scene-level function at all. This aligns with some participants' critical feedback on the reduced usefulness of the scene-level function (**T6-1**).

Quantitative Data Results ⟷ Qualitative Data Results

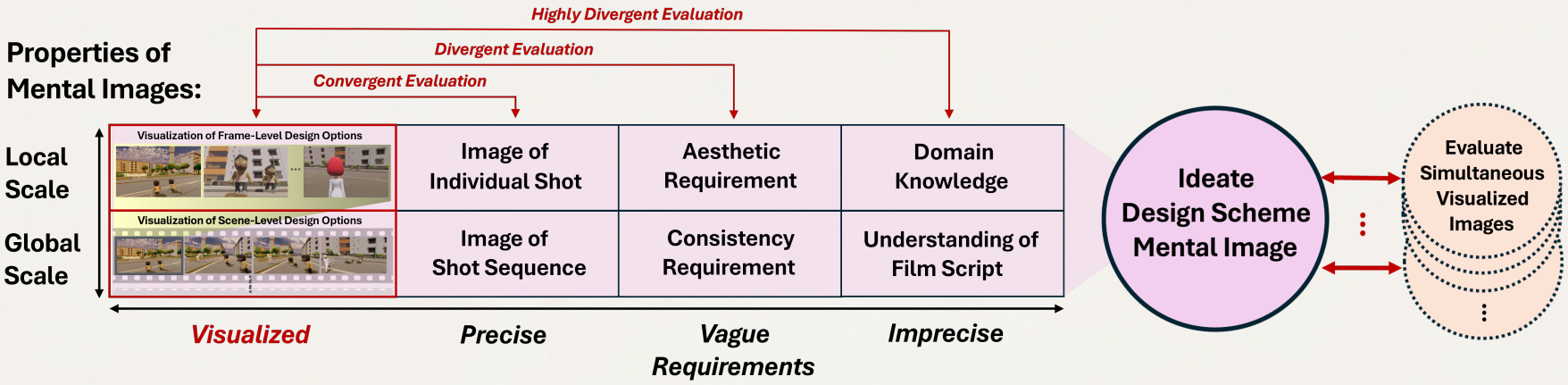
Compare, corroborate, and supplement

Key Finding of My Study

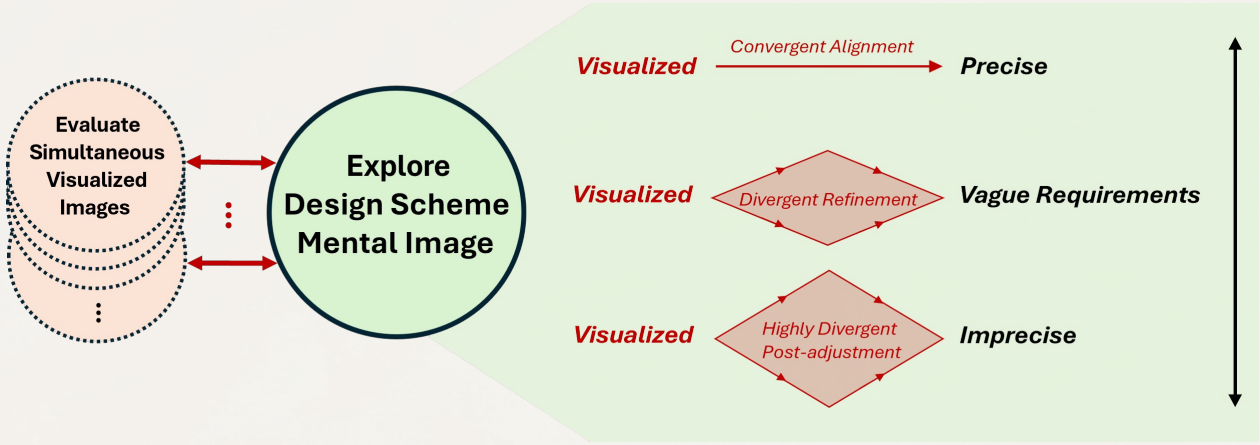
The *AISEE* Model: The AI-supported Simultaneous Evaluative and Exploratory Creative Thinking Model



Zooming in the human-AI creative cognition process of ***AISEE***...

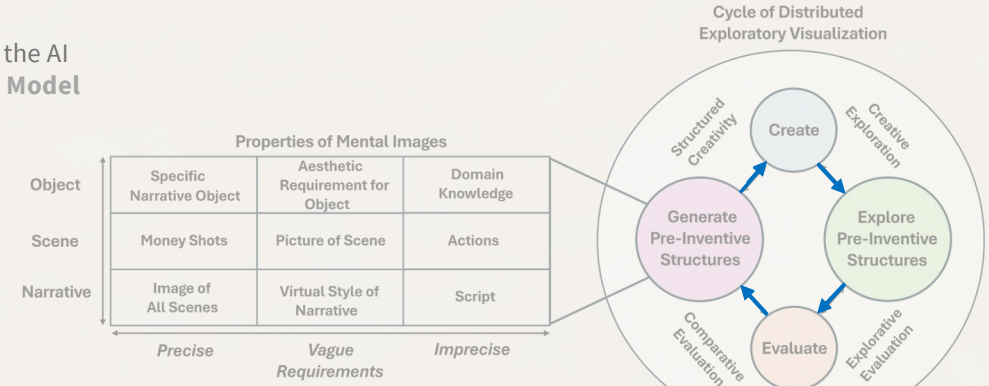


Zooming in the human-AI creative cognition process of ***AISEE***...

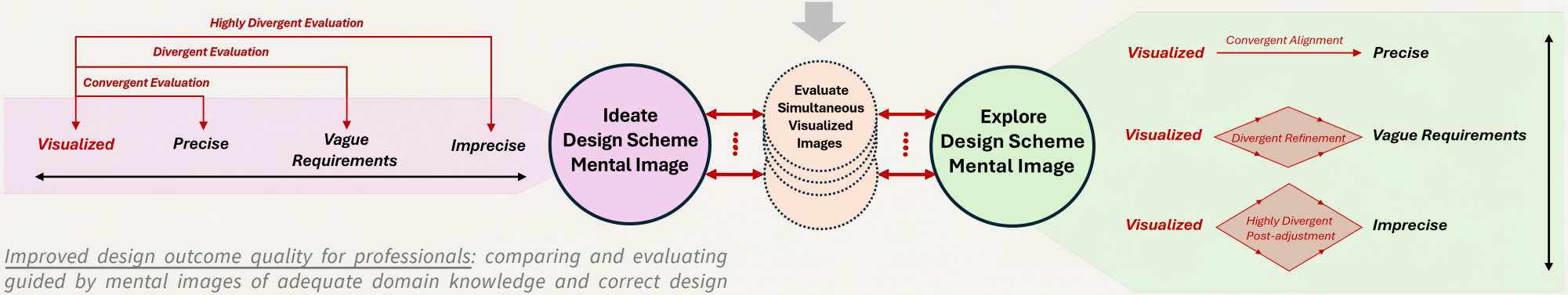


Creative cognition workflow comparison

Without assistance of the AI system: **Davis et al.'s Model** (Davis et al., 2011)



With assistance of the AI system: ***AISEE* Model**

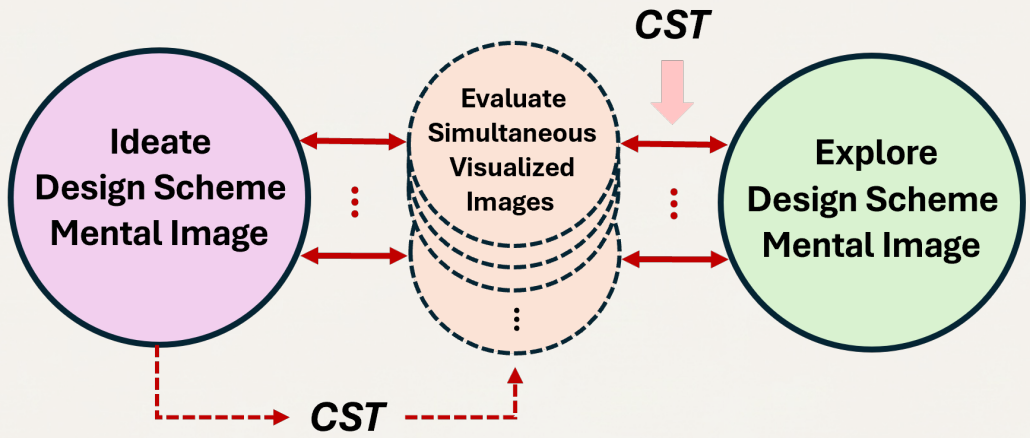


- *Improved design outcome quality for professionals: comparing and evaluating guided by mental images of adequate domain knowledge and correct design requirements*

- *Creative thinking efficiency enhanced for professionals: emphasizing convergent alignment and divergent refinement over highly divergent post-adjustment*

Latent Human Needs Revealed through ***AISEE***

Latent User Need 2: *Can concurrently explore design suggestions at a global design granularity level.*

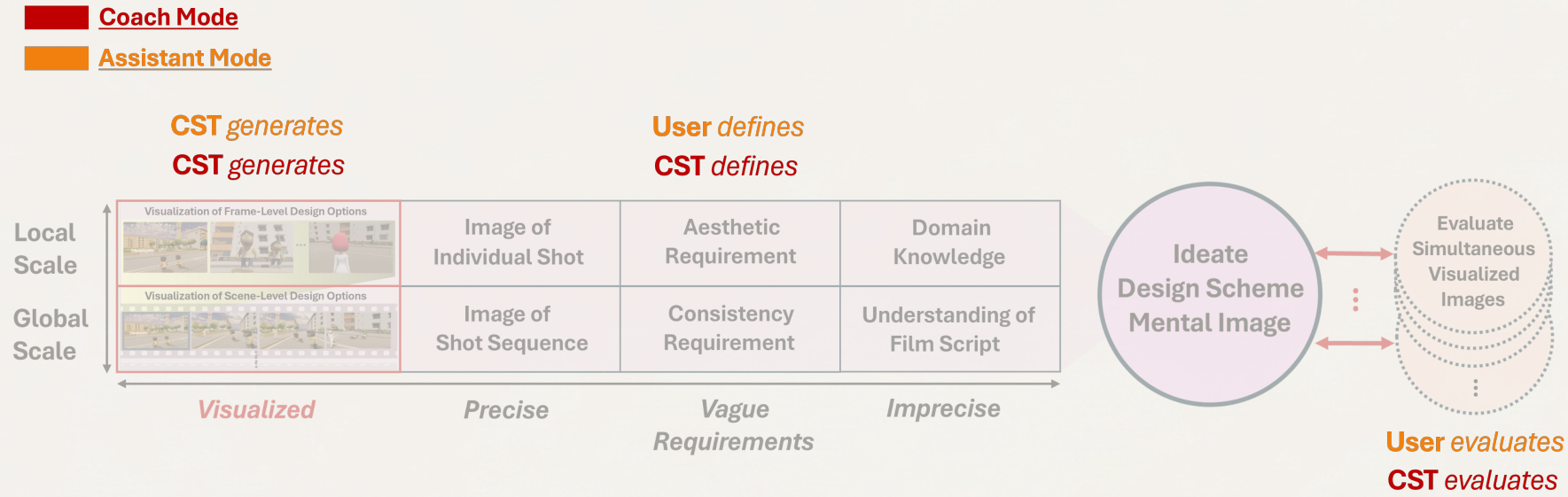


Latent User Need 1 *(from the users to the CST system): inform the system with their design requirements at different mental abstraction levels.*

Latent User Need 1 *(from the CST system to the users): can generate design suggestions that align with the users' requirements at varying abstraction levels.*

Design Implications for Building Effective Creativity Support Tool (CST)

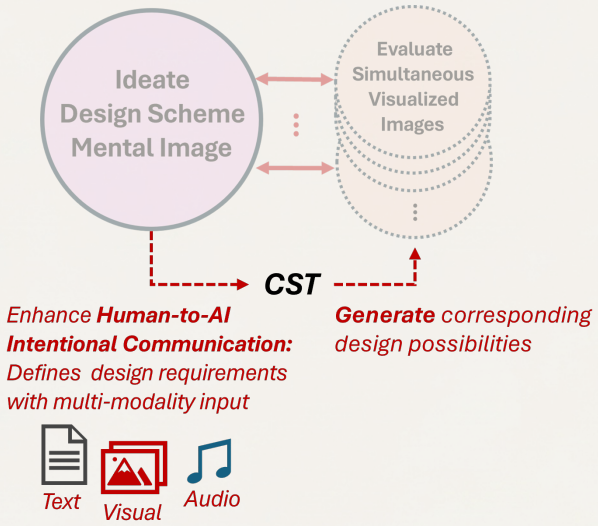
Design Implication 1: Resolving Domain Knowledge Disparities When Conducting Creative Tasks: Coach Mode and Assistant Mode



Makes mixed-initiative interaction (Horvitz, 1999) explicit and user-selectable, and further grounds it in the *AISEE* cognitive model by aligning guidance vs autonomy with different abstraction levels of creative intent rather than implicit adaptation.

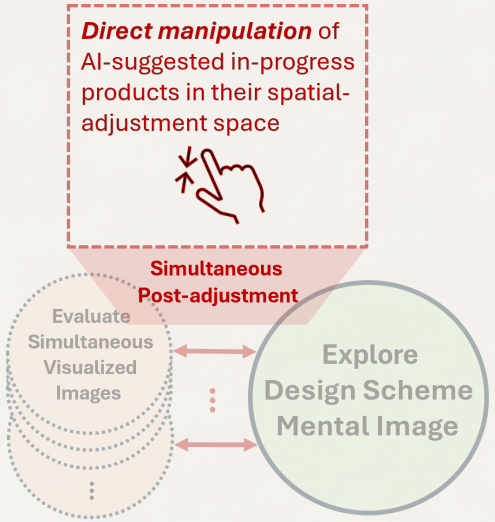
Design Implications for Building Effective Creativity Support Tool (CST)

Design Implication 2: Defining Design Requirements across Varying Mental Abstraction Levels: Multi-Modality Semantic Input



Extends CST support for multiple representations and ambiguity (Shneiderman, 2007; Buxton, 2007) by mapping modalities to distinction abstraction level of creative intent within human's cognitive workflow accounted by the *AISEE* model.

Design Implication 3: Situating AI Suggestions in Real-Time in the On-Screen 3D Spatial Adjustment Space and Enabling Direct Human Manipulative Adjustment



Advances direct manipulation (Shneiderman, 1983) and mixed-initiative creativity (Liapis et al., 2014) by embedding AI outputs as manipulable objects, and grounds this interaction in AISEE to tightly couple exploration and evaluation in real-time creative cognition.

Zimmerman, J., Forlizzi, J., & Evenson, S. (2007). Research through design as a method for interaction design research in HCI. In M. B. Rosson & D. J. Gilmore (Eds.), *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 493–502). Association for Computing Machinery. https://doi.org/10.1145/1240624.1240704

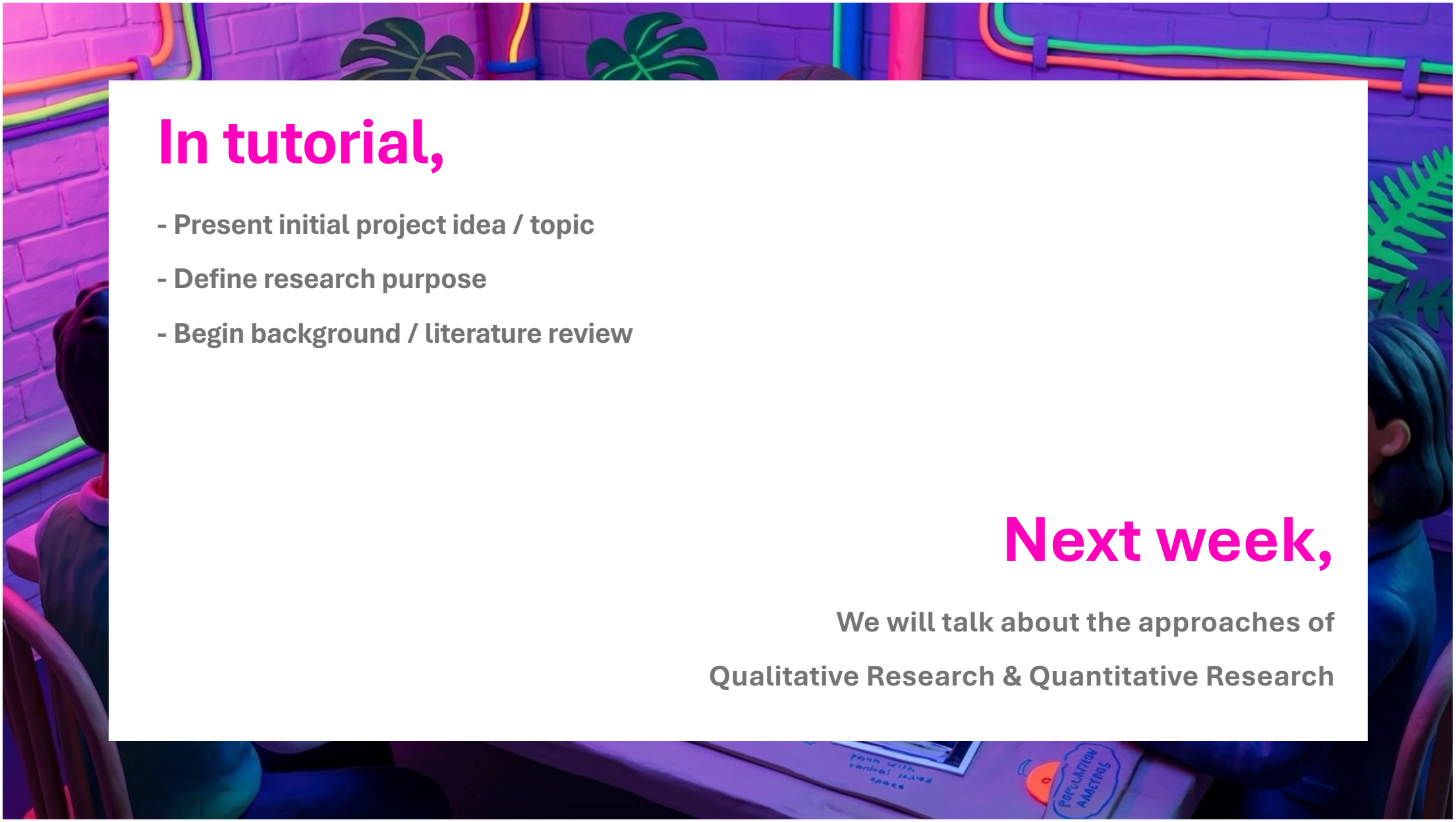
Hudson, S. E., & Mankoff, J. (2014). Concepts, values, and methods for technical human–computer interaction research. In J. S. Olson & W. A. Kellogg (Eds.), *Ways of knowing in HCI* (pp. 69–93). Springer. https://doi.org/10.1007/978-1-4939-0378-8_4



In today's class:

1. **Library Workshop**
2. **Lecture "What, Why, and Ethical Considerations"**
3. **Lecture "Research Through Design"**
4. **Specialism Studio Meeting with tutors**

V1014 (TED) | V1122 (ISD) | V1123 (IBD) | V1124 (SSD)



In tutorial,

- Present initial project idea / topic
- Define research purpose
- Begin background / literature review

Next week,

We will talk about the approaches of

Qualitative Research & Quantitative Research